

JSS 2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF JSS 1 WORK

Topic: Review of JSS1 Work

Content: Recap of farmland preparation and nursery practices; Discussion on importance of soil and crop selection.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define horticulture and crop production.
2. List the five main steps of farmland preparation in the correct order.
3. Define "nursery" and state two reasons for its importance.
4. Explain the importance of soil as a medium for plant growth.
5. List three factors to consider when selecting a crop to plant.

WHY WE FARM



Lesson Content:

1. Recap of Basic Definitions

- **Horticulture:** The art and science of cultivating garden crops, especially fruits, vegetables, flowers, and ornamental plants.
- **Crop Production:** The large-scale practice of growing agricultural crops (like maize, rice, cassava, yam) for food, feed, or industrial use.
- **Soil:** The upper layer of the earth's crust, which serves as a natural medium for plant growth. It provides anchorage, water, and nutrients.

2. Recap of Farmland Preparation This involves all the activities a farmer performs on the land *before* planting. The goal is to make the soil suitable for seed germination and plant growth.

- **A. Site Selection:** Choosing the best piece of land based on factors like:
 - **Topography:** The land should be flat or gently sloping to prevent erosion.
 - **Soil Type:** The soil must be suitable for the chosen crop (e.g., sandy-loam for vegetables).
 - **Water Access:** Must be near a reliable source of water (river, stream, well).
- **B. Land Clearing:** Removing unwanted vegetation (trees, bushes, grasses) from the site. This can be done manually (cutlass, axe) or mechanically (bulldozer).
- **C. Stumping:** The removal of tree stumps (the roots left in the ground after clearing). This is essential if the farmer plans to use machines like tractors.
- **D. Tillage (Ploughing, Harrowing):**
 - **Ploughing:** The first "breaking" of the soil to dig it up, loosen it, and bury weeds.
 - **Harrowing:** The second "breaking" of the soil to break the large soil clods into a finer, smoother, and more level seedbed.
- **E. Ridging/Bed Making:** Gathering the tilled soil into mounds (ridges) or flat-topped beds.
 - **Importance:** Provides deeper loose soil for root growth, controls erosion, and allows for easy water management.

3. Recap of Nursery Practices

- **Definition:** A nursery is a special, small plot of land where young plants (seedlings) are raised with extra care before they are moved to the permanent farm.
- **Crops that require a nursery:** Tomato, Pepper, Cabbage, Lettuce, Onion, Cocoa, Oil Palm.
- **Importance of a Nursery:**
 1. **Good Germination:** Provides the best conditions for tiny seeds to sprout.
 2. **Care of Seedlings:** It is easier to water, weed, and protect a small nursery than a large farm.
 3. **Selection of the Fittest:** Allows the farmer to select *only* the healthy, strong seedlings for transplanting, which ensures a good yield.
 4. **Saves Space:** The main farm can be used for other crops while the seedlings are growing in the nursery.

4. Importance of Soil and Crop Selection

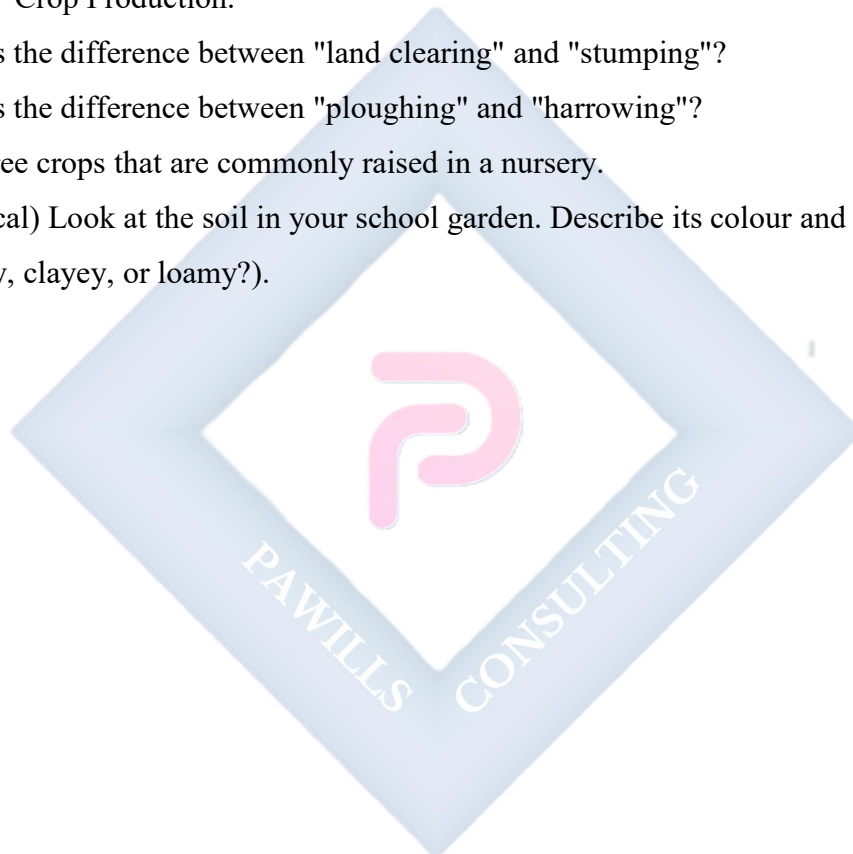
- **Why Soil is Important:**
 - **Anchorage:** Holds the plant's roots firmly in the ground.
 - **Nutrients:** Supplies all the essential minerals (Nitrogen, Phosphorus, Potassium) for growth.
 - **Water:** Holds the water that plants absorb through their roots.
 - **Air:** Provides oxygen for the roots to breathe.
- **Why Crop Selection is Important:**
 - **Climate:** You must plant crops that suit your climate (e.g., don't plant apples in the hot tropics).
 - **Soil Type:** Plant crops that match the soil (e.g., plant cassava in sandy soil, rice in clay/loamy soil).
 - **Economic Value:** Plant crops that are in high demand and will make a profit.
 - **Pest/Disease Resistance:** Choose crop varieties that are known to resist common diseases.

Evaluation:

1. What is horticulture?
2. List the five main steps of farmland preparation, in order.
3. What is a nursery?
4. State two reasons why a farmer would use a nursery.
5. List three reasons why soil is important for plant growth.

Assignment:

1. Define "Crop Production."
2. What is the difference between "land clearing" and "stumping"?
3. What is the difference between "ploughing" and "harrowing"?
4. List three crops that are commonly raised in a nursery.
5. (Practical) Look at the soil in your school garden. Describe its colour and texture (e.g., is it sandy, clayey, or loamy?).



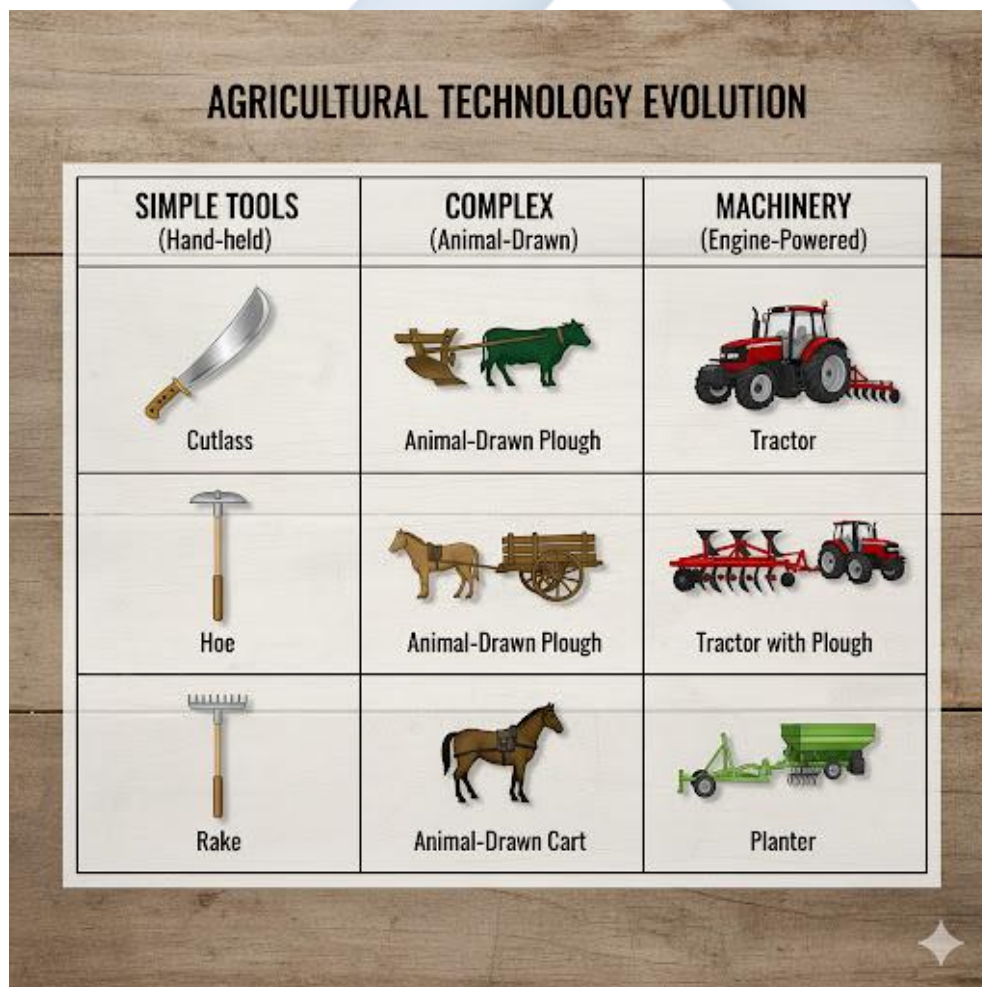
WEEK 2: FARM TOOLS AND MACHINES

Topic: Farm Tools and Machines

Content: Types and examples of farm machinery; Functions of simple and complex machines; Safety and maintenance practices.

Performance Objectives: By the end of this lesson, students should be able to:

1. Differentiate between simple farm tools, complex machines, and farm machinery.
2. List at least three examples of simple, complex, and machine-powered implements.
3. State the specific function of a tractor, plough, harrow, and planter.
4. List five general maintenance practices for farm tools and machines.
5. State five safety rules to observe when using farm machines.



Lesson Content:

1. Classification of Farm Equipment

- **A. Simple Farm Tools:**

- **Definition:** Simple hand-held tools that are *not* powered by animals or engines. They are powered by human energy.
- **Examples:** Cutlass (clearing), Hoe (weeding, making ridges), Rake (levelling, gathering trash), Shovel/Spade (lifting soil), Hand Fork (loosening soil), Hand Trowel (transplanting), Watering Can (watering).

- **B. Complex Machines (Animal-Drawn):**

- **Definition:** Implements that are more complex than hand tools and are designed to be *pulled* by farm animals (e.g., donkeys, oxen, camels).
- **Examples:** Animal-drawn Plough, Cart, Ridger.

- **C. Farm Machinery (Engine-Powered):**

- **Definition:** Heavy-duty, engine-powered machines that perform farm operations on a large, commercial scale.
- **The Tractor:** This is the *prime mover*. It is the "engine on wheels" that provides the power to *pull* or *operate* other machines (implements).
- **Tractor-Coupled Implements:**
 - **Plough:** Used for the *primary tillage* (first breaking) of hard soil.
 - **Harrow:** Used for *secondary tillage* (breaking soil clods into a fine seedbed).
 - **Ridger:** Used for making large ridges.
 - **Planter:** Used for planting seeds in rows automatically.
 - **Harvester:** Used for harvesting crops (e.g., combine harvester for maize/rice).
 - **Sprayer:** Used for the large-scale application of fertilizers, herbicides, or pesticides.

2. Functions of Key Farm Machinery

1. **Tractor:** Provides the pulling power (called "traction") and engine power (via a Power Take-Off or "PTO" shaft) to run other machines.
2. **Plough (Mouldboard Plough):** Its large curved blade cuts deep into the soil and *inverts* it (turns it upside down) to bury weeds and loosen the soil.
3. **Harrow (Disc Harrow):** A set of sharp, spinning discs that cut the large clods left by the plough into smaller, finer pieces, making the land smooth for planting.
4. **Planter (Seed Drill):** A machine that automatically:
 - Opens a small furrow (hole) in the soil.
 - "Drills" or drops seeds at a specific spacing.
 - Covers the seeds with soil.

3. Safety and Maintenance Practices

- **A. Safety Rules (Personal and Operational):**

1. **Training:** Never operate a machine (especially a tractor) without proper training.
2. **PPE (Personal Protective Equipment):** Always wear safety boots, gloves, and ear protection when operating loud machinery.
3. **No Passengers:** Never allow people to ride on the tractor or implements unless there is a proper seat.
4. **Moving Parts:** Keep hands, feet, and loose clothing *away* from all moving parts (belts, chains, shafts).
5. **Refuelling:** *Never* refuel a hot engine or while smoking.
6. **Guards:** Ensure all safety guards and shields are in place before starting.

- **B. Maintenance Practices (Keeping Tools in Good Condition):**

1. **Cleaning:** Clean all tools and machines *after each use* to remove soil and moisture.
2. **Sharpening:** Regularly sharpen the cutting edges of tools (cutlass, hoe, plough blades).
3. **Oiling/Greasing:** Apply oil or grease to all moving parts and metal surfaces to prevent rust and reduce friction.
4. **Proper Storage:** Store tools in a cool, dry, and secure tool shed.

5. **Check Engine (for machinery):** Regularly check the engine oil, water (coolant), and fuel levels before starting a tractor.

Evaluation:

1. What is the main difference between a simple tool and farm machinery?
2. What is the name of the machine used for primary tillage (first breaking of soil)?
3. What is the name of the machine used for secondary tillage (breaking clods)?
4. List three safety rules for using farm machinery.
5. List three maintenance practices for farm tools.

Assignment:

1. Draw and label a Cutlass, a Rake, and a Hand Trowel.
2. What is the primary function of a Tractor?
3. What is the difference between a Planter and a Harvester?
4. Why is it important to clean your tools after every use?
5. What does "PPE" stand for, and give two examples used on a farm.

WEEK 3: SOIL FERTILITY AND IMPROVEMENT

Topic: Soil Fertility and Improvement **Content:** Meaning of soil fertility; Causes of loss of fertility; Methods of improving soil fertility (organic/inorganic); Compost making.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "soil fertility."
2. List at least five essential plant nutrients.
3. Explain three ways soil fertility is lost.
4. Differentiate between organic manures and inorganic (chemical) fertilizers.
5. Describe the process of making compost.

SOIL FERTILITY

CAUSES OF FERTILITY LOSS



Leaching



Erosion



Over-cropping

METHODS OF IMPROVEMENT



Organic
(Compost, Manure)



Inorganic
(Fertilizers)



Crop Rotation

Lesson Content:

1. Meaning of Soil Fertility

- **Definition:** Soil fertility is the *ability of the soil* to provide all the essential nutrients in the right amounts and proportions, and support plant growth to produce a high yield.
- A fertile soil is *not* just "rich in nutrients." It must also have:
 - Good texture and structure (e.g., loamy)
 - Good water-holding capacity.
 - Good aeration (air spaces for roots).
 - A healthy population of soil organisms (like earthworms and bacteria).
- **Plant Nutrients:** Plants are like humans; they need food. These "foods" are called nutrients.
 - **Macronutrients (needed in large amounts):** Nitrogen (N), Phosphorus (P), Potassium (K).
 - **Secondary Nutrients:** Calcium (Ca), Magnesium (Mg), Sulphur (S).
 - **Micronutrients (needed in tiny amounts):** Iron (Fe), Zinc (Zn), Boron (B), etc.

2. Causes of Loss of Soil Fertility

Soil can "lose its power" or become infertile in several ways:

1. **Leaching:** This is the "washing down" of soluble nutrients (especially Nitrogen) deep into the soil by heavy rainfall, moving them beyond the reach of plant roots.
2. **Soil Erosion:** The "washing away" or "blowing away" of the fertile *topsoil* by water or wind. This is the worst form of fertility loss.
3. **Over-cropping / Continuous Cropping:** Planting the *same crop* on the *same land* year after year. The crop will "use up" all of its specific favourite nutrients (e.g., maize uses a lot of Nitrogen).
4. **Overgrazing:** Allowing too many animals (cows, goats) to graze on a piece of land. They eat all the grass, leaving the soil bare and exposed to erosion.
5. **Bush Burning:** Burning the farm after clearing. This kills valuable soil micro-organisms and releases nutrients (like Nitrogen) into the air as gas.

3. Methods of Improving Soil Fertility

We can "feed" the soil to make it fertile again.

- **A. Organic Manures (Natural):**
 - **Definition:** Fertilizers made from natural, living (or once-living) materials. They improve *both* nutrient content and soil *structure*.
 - **Examples:**
 - **Animal Manure:** (Poultry droppings, cow dung, goat manure).
 - **Green Manure:** Planting specific leafy plants (like legumes) and then ploughing them into the soil while they are still green.
 - **Compost:** Decomposed (rotted) mixture of plants and animal waste.
- **B. Inorganic/Chemical Fertilizers (Artificial):**
 - **Definition:** Man-made chemicals that are produced in a factory to supply specific nutrients. They are fast-acting but *do not* improve soil structure.
 - **Examples:**
 - **NPK 15:15:15:** A "complete" fertilizer containing Nitrogen, Phosphorus, and Potassium.
 - **Urea:** Provides only Nitrogen.
 - **SSP (Single Superphosphate):** Provides only Phosphorus.
- **C. Cultural Practices:**
 - **Crop Rotation:** Planting *different* types of crops on the same land in a specific order (e.g., Maize -> Beans -> Yam). This helps to replace nutrients and control pests.
 - **Cover Cropping:** Planting "cover crops" (like legumes) to cover the soil and prevent erosion.

4. How to Make Compost (The Indore Method)

- **Definition:** Compost is a dark, crumbly, nutrient-rich soil-like material made from the *decomposition* (rotting) of organic waste.
- **Materials:**
 - **Greens (Nitrogen-rich):** Kitchen waste (fruit/veg scraps), fresh grass clippings, animal manure.
 - **Browns (Carbon-rich):** Dry leaves, dry grass, paper, small twigs.
 - **Other:** Water, soil (to add micro-organisms).

- **Process:**

1. **Site:** Choose a shaded, well-drained spot.
2. **Layer 1:** Start with a 15cm layer of "Browns" (twigs, dry leaves) for air flow.
3. **Layer 2:** Add a 15cm layer of "Greens" (kitchen waste, fresh grass).
4. **Layer 3:** Add a thin layer of "Browns" (dry leaves).
5. **Layer 4:** Add a thin layer of animal manure or soil (this adds the bacteria to start the decomposition).
6. **Water:** Sprinkle water on the pile (it should be damp, not soaking).
7. **Repeat:** Repeat layers (Browns -> Greens -> Manure/Soil -> Water) until the pile is about 1 meter high.
8. **Turn:** Use a shovel to *turn* or *mix* the pile every 2-3 weeks. This provides the oxygen needed for fast decomposition.
9. **Ready:** The compost is ready in 2-4 months when it is dark, crumbly, and smells like fresh earth.

Evaluation:

1. What is "soil fertility"?
2. What are the three "macronutrients" (N, P, K)?
3. Explain two ways soil can lose its fertility.
4. What is the difference between organic manure and inorganic fertilizer?
5. What are the "Greens" and "Browns" used for in compost making?

Assignment:

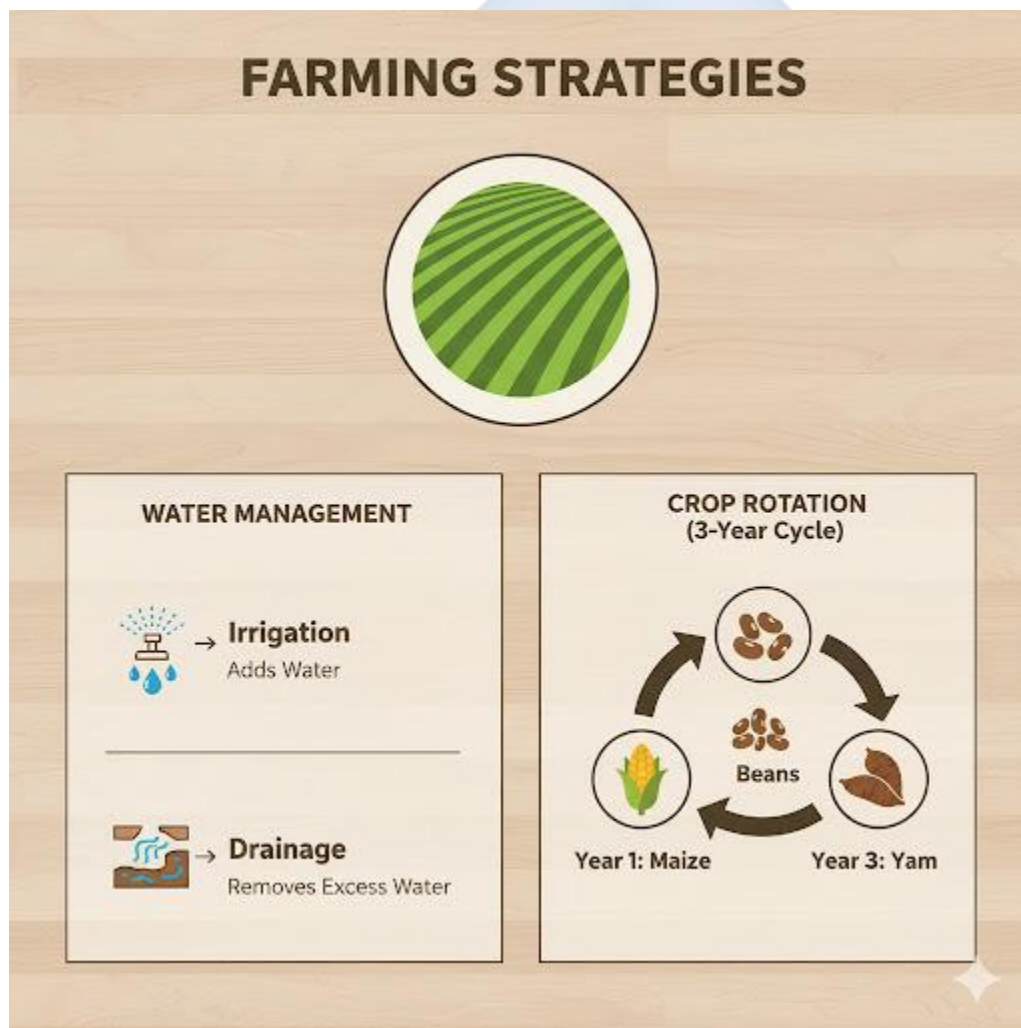
1. Define Leaching, Soil Erosion, and Over-cropping.
2. What is "green manure"?
3. What is "NPK 15:15:15"?
4. Why is it important to "turn" a compost pile?
5. (Practical) Start a small compost pile or bucket in the school garden. Write down the (1) Greens and (2) Browns that you added to it.

WEEK 4: LAND PREPARATION TECHNIQUES

Content: Clearing, tilling, and leveling; Irrigation and drainage systems; Crop rotation practices.

Performance Objectives: By the end of this lesson, students should be able to:

1. Review the steps of land preparation (clearing, tilling, levelling).
2. Define "irrigation" and list two common methods.
3. Define "drainage" and state its importance.
4. Define "crop rotation" and give an example of a 3-course rotation.
5. Explain two reasons why crop rotation is beneficial.



Lesson Content:

1. Review of Land Preparation (from Week 1)

- **Clearing:** Removing vegetation (bushes, trees) using cutlasses, axes, or bulldozers.
- **Tillage:** The mechanical "working" of the soil to prepare it for planting.
 - **Primary Tillage (Ploughing):** Deep breaking of the soil.
 - **Secondary Tillage (Harrowing):** Breaking soil clods into a fine, level seedbed.
- **Levelling:** Making the soil surface flat and smooth to ensure uniform planting and water distribution.
- **Ridging/Bed Making:** Forming the soil into raised mounds (ridges) or beds.

2. Water Management: Irrigation and Drainage Water is essential, but *too much* or *too little* is bad.

- **A. Irrigation:**
 - **Definition:** The *artificial application* of water to farmland in order to supply crops with the water they need during periods of low rainfall (the dry season).
 - **Methods:**
 1. **Surface Irrigation (Flooding):** Allowing water to flow over the surface of the land. (Common in rice paddies).
 2. **Sprinkler Irrigation:** Spraying water over the crops like "artificial rain," using sprinklers.
 3. **Drip/Trickle Irrigation:** Using pipes with small holes (emitters) that *drip* water slowly, directly onto the base of each plant. This is the most efficient method as it wastes very little water.
 4. **Manual Irrigation:** Using a watering can (for small gardens/nurseries).
- **B. Drainage:**
 - **Definition:** The *removal* of excess or stagnant water (waterlogging) from farmland.
 - **Importance:** Most plant roots cannot survive in waterlogged soil because they cannot get *oxygen* and will rot.

- **Methods:**

1. **Surface Drainage:** Digging *open ditches* or *channels* (furrows) to allow excess water to flow away by gravity.
2. **Sub-Surface Drainage:** (Advanced) Placing perforated pipes *underground* to collect and drain the water.

3. Crop Rotation Practices

- **Definition:** Crop rotation is the practice of planting *different types* of crops in a *planned sequence* on the *same piece of land* over several years.
- **It is the *opposite* of continuous cropping** (planting maize on the same land every single year).
- **Principle:** Different crops have different needs and effects on the soil.
 - **Legumes (e.g., Beans, Cowpea, Groundnut):** *Add* Nitrogen to the soil (via root nodules).
 - **Cereals (e.g., Maize, Rice):** *Use a lot* of Nitrogen.
 - **Root Crops (e.g., Yam, Cassava):** *Use a lot* of Potassium and open up the soil.
- **Example of a 3-Course Rotation (3 Years):**
 - **Year 1:** Plant **Maize** (uses lots of Nitrogen).
 - **Year 2:** Plant **Beans** (adds Nitrogen back to the soil).
 - **Year 3:** Plant **Yam** (uses Potassium and benefits from the Nitrogen left by the beans).
 - **Year 4:** Repeat the cycle (back to Maize).

4. Benefits of Crop Rotation

1. **Soil Fertility Improvement:** The rotation (especially with legumes) naturally replaces soil nutrients, reducing the need for expensive chemical fertilizers.
2. **Pest and Disease Control:** Many pests and diseases attack *only one* type of crop. By changing the crop, the pest's life cycle is broken (they starve or die off).
3. **Weed Control:** Different crops have different planting times and canopies, which helps to disrupt the growth cycles of specific weeds.

4. **Improved Soil Structure:** Rotating deep-rooted crops (like yam) with shallow-rooted crops (like vegetables) helps to improve soil structure and water absorption.

Evaluation:

1. What is the difference between primary tillage and secondary tillage?
2. Define "irrigation."
3. What is "drip irrigation," and why is it efficient?
4. What is "drainage," and why is it important?
5. What is "crop rotation"?

Assignment:

1. Define Irrigation, Drainage, and Crop Rotation.
2. List two methods of irrigation and one method of drainage.
3. Give two *major* benefits of practicing crop rotation.
4. Design a simple 3-course crop rotation for a farm, using crops common in your area.
5. Why is it *bad* to plant maize on the same piece of land every single year? (This is continuous cropping).

WEEK 6: SITE SELECTION FOR CROP PRODUCTION

Topic: Site Selection for Crop Production

Content: Factors affecting crop site selection; Land use and suitability; Environmental and economic factors.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "site selection."
2. List three climatic factors to consider during site selection.
3. List three soil (edaphic) factors to consider.
4. List three economic factors to consider.
5. Explain how "land use" affects site selection.



Lesson Content:

1. What is Site Selection?

- **Definition:** Site selection is the *critical process* of evaluating and choosing the *best possible piece of land* on which to establish a farm for growing specific crops.
- **Importance:** A good site = high yield and high profit. A bad site = low yield, crop failure, and loss of money.

2. Factors Affecting Crop Site Selection

There are three main categories of factors to consider:

- **A. Climatic Factors (The Weather):**
 1. **Rainfall:** Does the area receive enough rainfall for the crop? Is the rainfall reliable? If not, is there a source for irrigation?
 2. **Temperature:** Is the area too hot or too cold for the crop? (e.g., Cabbage prefers cooler temperatures, while yams need heat).
 3. **Sunlight:** Does the land receive at least 6-8 hours of direct sunlight per day? (Most crops need full sun). Is the land shaded by mountains or large buildings?
 4. **Wind:** Is the area prone to very strong winds that could damage crops?
- **B. Soil (Edaphic) Factors:**
 1. **Soil Type (Texture):** Is the soil sandy (drains too fast), clayey (holds too much water), or loamy (the best, a balanced mix)?
 2. **Soil Fertility:** Is the soil rich in nutrients? (A soil test can determine this).
 3. **Soil Depth:** Is the soil deep enough for roots to grow? (e.g., You can't grow yams on shallow, rocky soil).
 4. **Topography (Slope):** The land should be flat or *gently sloping*. Very *steep* land is bad because it leads to rapid soil erosion.
- **C. Biotic Factors:**
 1. **Pests and Diseases:** Is the area known to have a high population of specific pests (like stem borers) or diseases (like cassava mosaic virus)?
 2. **Weeds:** Is the land covered in very difficult-to-remove weeds (like spear grass)?

- **D. Economic and Social Factors (The "Business" Factors):**

1. **Proximity to Market:** How far is the farm from the market where you will sell your crops? (If it's too far, transport costs will "eat" all your profit).
2. **Transportation:** Is there a good road to the farm, or is it a bush path?
3. **Labour:** Is there a good supply of skilled farm workers in the local village or town?
4. **Security:** Is the area safe from thieves (who steal crops) or stray animals (cows, goats) that can destroy the farm?

3. Land Use and Suitability

- **Land Use:** What is the land *currently* being used for?
- **Suitability:** The farmer must check if the previous land use has damaged the land (e.t., if it was a mechanic workshop, the soil might be polluted with oil and unsuitable for vegetables). The land must be *suitable* for agriculture.
- Government policies (e.g., "This land is for residential use only") can also affect site selection.

Evaluation:

1. What is "site selection"?
2. List two climatic factors and two soil factors that affect site selection.
3. What does "topography" mean, and why is it important?
4. Why is "proximity to market" an important *economic* factor?
5. Why is a very steep slope bad for farming?

Assignment:

1. List the four main categories of factors for site selection.
2. Explain "proximity to market" and "labour supply" as economic factors.
3. What is a "soil test," and how can it help in site selection?
4. (Practical) Imagine you have to start a school vegetable garden. Walk around your school.
 - (a) Choose the *best* site.

- (b) Choose the *worst* site.
5. For your *best* site, explain *why* you chose it, using 3 factors from the lesson (e.g., "I chose it because it has good sunlight, it is flat, and it is near the water tap.").



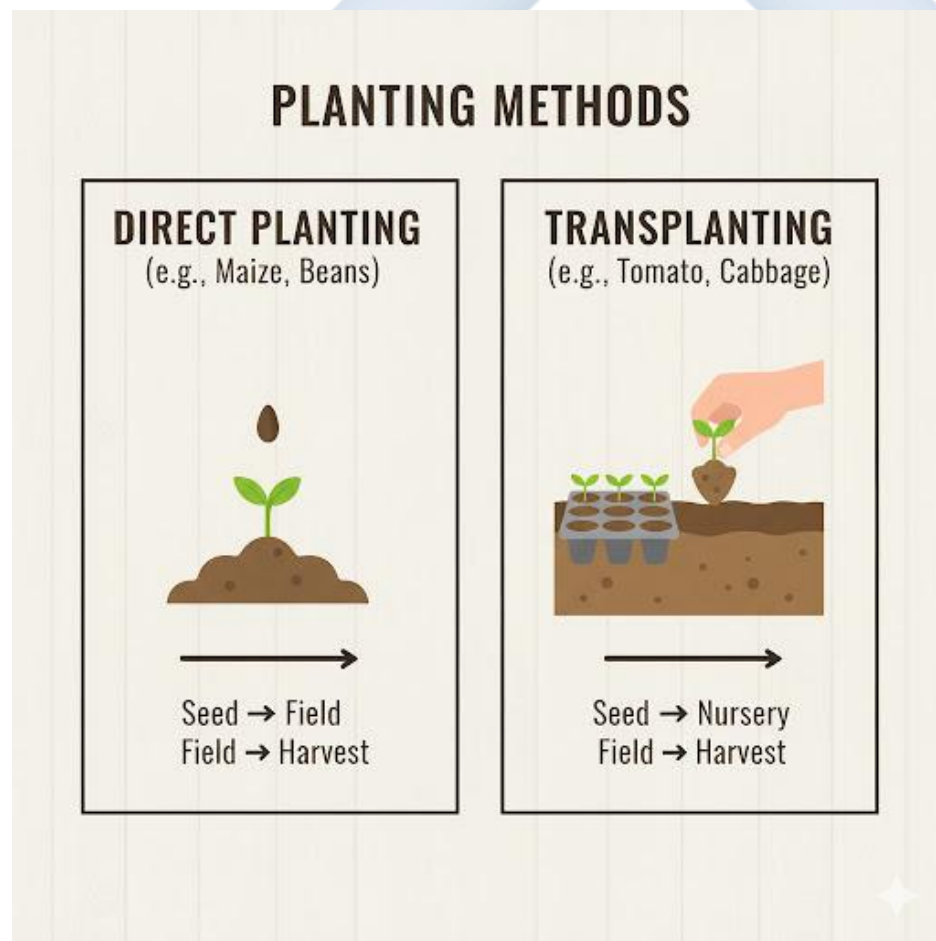
WEEK 8: CROP SPACING AND PLANTING METHODS

Topic: Crop Spacing and Planting Methods

Content: Meaning and importance of crop spacing; Methods of planting (direct, transplanting); Plant population and density.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "crop spacing" and "plant population."
2. State three reasons why correct crop spacing is important.
3. Differentiate between "direct planting" and "transplanting."
4. Give two examples of crops for each planting method.
5. Explain the relationship between spacing and plant population.



Lesson Content:

1. Meaning of Crop Spacing

- **Crop Spacing:** This is the *specific distance* kept between one plant and another on a farm, both *along the row* (intra-row) and *between the rows* (inter-row).
- **Examples:**
 - Maize: 75cm *between* rows, 25cm *along* the row.
 - Yam: 1m x 1m (1m *between* rows, 1m *along* the row).
- The spacing is different for every crop.

2. Importance of Correct Crop Spacing

Giving plants the "correct space" is crucial.

- **If spacing is too wide:**
 1. **Wastes Land:** You will have a low plant population and a low total yield.
 2. **Encourages Weeds:** The open, sunny spaces will be quickly covered by weeds.
- **If spacing is too close (overcrowding):**
 1. **Competition:** Plants will "fight" each other for limited resources (sunlight, water, nutrients).
 2. **Weak, Lanky Plants:** Plants will grow very tall and thin (lanky) trying to reach for sunlight, and will be weak.
 3. **Low Yield:** The individual plants will be stressed and produce very small fruits or seeds.
 4. **Spreads Disease:** Air cannot move freely between plants, so fungal diseases spread easily in the damp, crowded conditions.
- **Correct spacing** ensures that every plant gets enough resources to grow strong and produce a maximum yield.

3. Plant Population and Density

- **Plant Population:** The *total number* of plants in a given area (e.g., one hectare).

- **Formula:** $\text{Plant Population} = (\text{Total Farm Area}) / (\text{Spacing Area})$
 - *Example:* How many maize plants in a 100m x 100m (10,000 m²) field, using a 1m x 1m spacing?
 - $\text{Plant Population} = 10,000 \text{ m}^2 / (1\text{m} \times 1\text{m}) = 10,000 \text{ plants.}$
- **Plant Density:** How crowded the plants are. This is controlled by spacing.

4. Methods of Planting

There are two main ways to put a plant in the field.

- **A. Direct Planting (or Direct Sowing):**
 - **Definition:** This is when the *seeds* are planted *directly* into the permanent farmland where they will grow and mature.
 - **Methods:**
 - **Broadcasting:** Scattering the seeds randomly by hand (e.g., rice, some vegetables). It is fast but wasteful.
 - **Drilling:** Planting the seeds in a straight, continuous line (furrow).
 - **Row Planting:** Planting the seeds at specific, spaced-out positions along a row (the best method).
 - **Crops:** Maize, Cowpea (Beans), Yam, Cassava, Groundnut, Okra.
- **B. Transplanting:**
 - **Definition:** This is a *two-stage* process. The seeds are first planted in a **nursery** (see Week 1). After the seedlings are strong (e.g., 3-4 weeks), they are carefully *uprooted* and *transplanted* (moved) to the permanent farm.
 - **Crops:** Tomato, Pepper, Cabbage, Onion, Lettuce, Rice (often).
- **Hardening Off:** The process of gradually *acclimatizing* nursery seedlings to the harsh conditions of the open field (e.g., by reducing watering) *before* transplanting. This reduces "transplanting shock."

Evaluation:

1. What is "crop spacing"?
2. Give two reasons why "overcrowding" (spacing too close) is bad for crops.
3. What is "direct planting"?

4. What is "transplanting"?
5. Give one example of a crop that is transplanted.

Assignment:

1. Define Plant Population and Plant Density.
2. A farmer plants yams with a spacing of 1m x 1m. How many yam plants can fit on a piece of land that is 10m x 10m? (Show your calculation).
3. What is the main advantage of "transplanting" over "direct planting"? (Hint: Nursery).
4. What is "broadcasting," and what is its main disadvantage?
5. What is "hardening off," and why is it done?



WEEK 9: WEED MANAGEMENT

Topic: Weed Management

Content: Meaning and types of weeds; Effects of weeds; Methods of weed control (manual, chemical, biological).

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "weed" in an agricultural context.
2. List three common examples of weeds.
3. Explain three harmful effects of weeds on crops.
4. Differentiate between manual, chemical, and biological weed control.
5. Give one example of each control method.

WEED CONTROL METHODS



MANUAL:
Hand-pulling, Hoeing



CHEMICAL:
Herbicides (Pre/Post-emergence)



BIOLOGICAL:
Natural Enemies



CULTURAL:
Mulching, Good Spacing

Lesson Content:

1. Meaning and Types of Weeds

- **Definition:** A **weed** is any plant growing where it is *not wanted*.
 - A maize plant in a yam farm is a weed.
 - A tomato plant in a rice paddy is a weed.
- Weeds are plants that are very successful at competing with farm crops.
- **Common Examples:**
 - **Spear Grass** (*Imperata cylindrica*)
 - **Goat Weed** (*Ageratum conyzoides*)
 - **Pigweed** (*Amaranthus* species)
 - **Water Leaf** (*Talinum triangulare*)
 - **Tridax** (*Tridax procumbens*)

2. Harmful Effects of Weeds (Why Weeds are Bad) Weeds are a major problem for farmers because they reduce crop yield.

1. **Competition:** Weeds *compete* with crops for all essential resources:
 - Sunlight
 - Water
 - Soil Nutrients
 - Space
 - (Weeds are usually *better* at competing than the crops).
2. **Alternate Hosts:** Weeds can serve as *alternative hosts* (a "hiding place") for crop pests and diseases. (e.g., Insects hide in the weeds and then attack the crops).
3. **Reduced Crop Quality:** The presence of weed seeds in harvested grain (like rice) reduces its market value and quality.
4. **Block Farm Operations:** Weeds can block irrigation channels and make harvesting difficult.

3. Methods of Weed Control

- **A. Manual / Mechanical Control (Physical):**
 - **Definition:** Using physical labour or tools to remove or destroy weeds.
 - **Methods:**
 - **Hand-Pulling:** Physically pulling the weeds out by hand. (Effective in gardens/nurseries).
 - **Hoeing:** Using a simple hoe to cut the weeds. (Most common method in Nigeria).
 - **Slashing:** Using a cutlass to cut the weeds down.
 - **Mulching:** Covering the soil surface with a layer of material (like dry grass, wood chips, or plastic sheets) to *block sunlight* so weed seeds cannot germinate.
- **B. Chemical Control:**
 - **Definition:** Using chemicals called **herbicides** to kill weeds.
 - **Types of Herbicides:**
 - **Pre-emergence:** Applied to the soil *before* the weeds or crops sprout.
 - **Post-emergence:** Applied to the leaves of the weeds *after* they have started growing.
 - **Advantages:** Fast, effective on large farms, reduces labour.
 - **Disadvantages:** Can be expensive, can pollute the environment, requires training to use safely.
- **C. Biological Control:**
 - **Definition:** Using a weed's *natural enemies* (living organisms) to control its population.
 - **This is *not* about using cows or goats**, as they will eat the crops too!
 - **Method:** Scientists find a *specific* insect or disease that *only* eats the target weed and *does not* eat the crop. They then release this insect into the farm.
 - **Advantage:** Very safe for the environment, long-lasting.
 - **Disadvantage:** Very slow, very complex to research.
- **D. Cultural Control:**
 - Using good farming practices to *prevent* weeds from growing.

- **Example:** Planting at the *correct spacing* (see Week 8) so the crop leaves (canopy) quickly cover the ground and *shade out* the weeds.

Evaluation:

1. What is a weed?
2. Give one common example of a weed.
3. List three ways weeds are harmful to crops.
4. What is the difference between manual and chemical weed control?
5. What is a "herbicide"?

Assignment:

1. Define Weed, Herbicide, and Biological Control.
2. What is "competition" in farming?
3. What is "mulching," and how does it control weeds?
4. What is the main *disadvantage* of using chemical herbicides?
5. (Practical) Look around your school compound. Find and name one plant that you think is a "weed."

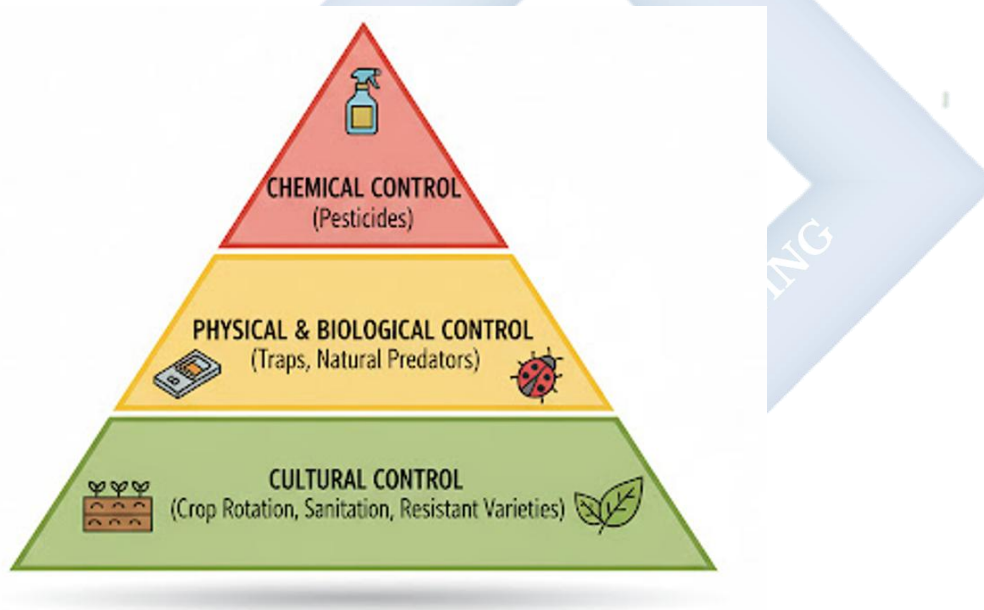
WEEK 10: CROP PROTECTION

Topic: Crop Protection

Content: Pests and diseases of crops; Control measures; Integrated pest management.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "pest" and "disease" in an agricultural context.
2. Give an example of an insect pest and a disease.
3. List three common control measures for pests.
4. Differentiate between pesticides, insecticides, and fungicides.
5. Define "Integrated Pest Management (IPM)."



Integrated Pest Management (IPM) Pyramid)

Lesson Content:

1. Pests and Diseases of Crops

- **A. Crop Pests:**

- **Definition:** A **pest** is any animal or insect that attacks or causes damage to crops, reducing the yield or quality.
- **Examples:**
 - **Insects:** (Most common)
 - *Grasshoppers/Locusts:* Eat leaves.
 - *Stem Borers:* (e.g., in maize) Larvae that bore into the stem.
 - *Aphids/Whiteflies:* Suck the sap from leaves.
 - *Weevils:* (e.g., in beans, rice) Bore into stored grains.
 - **Vertebrates:**
 - *Birds:* Eat seeds and fruits.
 - *Rodents (Rats, Rabbits):* Eat seedlings, yams, cassava.
- **B. Crop Diseases:**
 - **Definition:** A **disease** is a disorder caused by a pathogen (a micro-organism) that disrupts the normal growth of a plant.
 - **Pathogens (Causes of Disease):**
 - **Fungi:** (Most common) Cause "blight," "rust," "mildew," and "rot."
 - **Bacteria:** Cause "wilts" and "spots."
 - **Viruses:** (e.g., Cassava Mosaic Virus, Tomato Mosaic Virus) Cause yellowing, stunted growth.
 - **Signs:** Spots on leaves, rotted fruit, wilting (dying) plants, stunted growth.

2. Control Measures

- **A. Physical/Mechanical Control:**
 - **Hand-picking:** Physically picking large pests (like caterpillars) off plants.
 - **Traps:** Using sticky traps for insects or rat traps for rodents.
 - **Fences:** Building fences to keep out vertebrate pests (goats, rats).
 - **Scarecrows:** To scare away birds.
- **B. Cultural Control:**
 - Using good farm practices to *prevent* pests.
 - **Crop Rotation (Week 4):** Breaks the life cycle of pests.
 - **Proper Sanitation:** Removing and burning diseased plants.

- **Resistant Varieties:** Planting crop types that are naturally resistant to certain diseases.
- **C. Biological Control:**
 - Using a pest's *natural enemy* to control it (e.g., using ladybugs to eat aphids).
- **D. Chemical Control (Pesticides):**
 - **Definition:** Using **pesticides** (chemicals) to kill pests and pathogens.
 - **Types:**
 - **Insecticides:** Kill insects.
 - **Fungicides:** Kill fungi.
 - **Rodenticides:** Kill rodents (rats).
 - **Herbicides (Week 9):** Kill weeds.
 - **Disadvantages:** Can kill beneficial insects (like bees), pollute water, and be harmful to the farmer if not used safely.

3. Integrated Pest Management (IPM)

- **Definition:** IPM is a *smart* and *sustainable* approach to pest control.
- **Concept:** Instead of *only* using chemicals, IPM *combines* multiple control methods (cultural, biological, and physical) to keep pests at a low, acceptable level.
- **Process:**
 1. A farmer *monitors* the farm.
 2. They use *cultural* methods (like crop rotation) first.
 3. If pests appear, they try *physical* methods (like traps).
 4. Chemical pesticides are used *only as a last resort* when pests reach a level that will cause economic damage.
- **Goal:** To control pests effectively while being safe for the environment and the farmer.

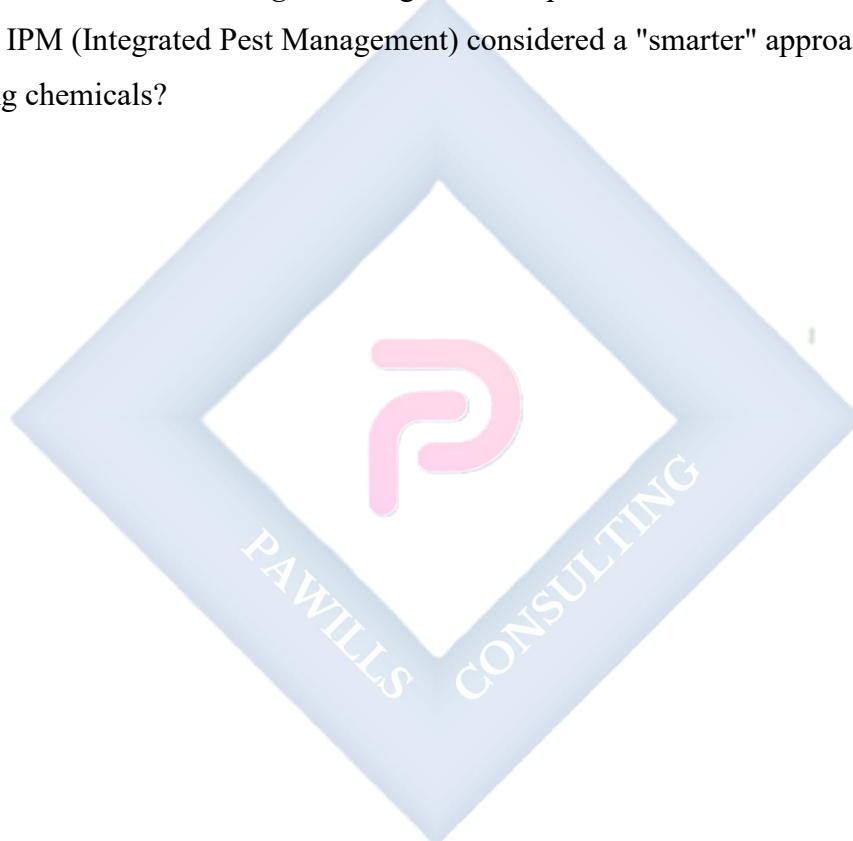
Evaluation:

1. What is a "crop pest"?
2. What is a "pathogen"?
3. What is the difference between an insecticide and a fungicide?

4. Give one example of "cultural control."
5. What is "Integrated Pest Management (IPM)"?

Assignment:

1. Define Pest, Disease, and Pathogen.
2. List three types of pathogens that cause plant diseases.
3. List three types of insect pests and the damage they do.
4. What is the main *disadvantage* of using chemical pesticides?
5. Why is IPM (Integrated Pest Management) considered a "smarter" approach than just spraying chemicals?



WEEK 11: REVISION

Topic: Revision

Content: Review of all topics in Term 1

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term (e.g., Soil Fertility, Irrigation, Crop Rotation, Weed, Pest, IPM).
2. Explain the difference between organic manures and inorganic fertilizers.
3. Explain the importance of crop spacing.
4. List the three main methods of weed and pest control (manual, chemical, biological).
5. Answer practice questions covering all topics from the term.

Lesson Content:

1. Summary of Term 1 Topics

- **Week 1 (Review):** We remembered the basics: Farmland preparation (Clearing, Stumping, Tilling), Nursery practices, and the importance of soil.
- **Week 2 (Tools):** We classified tools (Simple, Complex, Machinery). We learned the functions of the **Tractor** (prime mover), **Plough** (primary tillage), **Harrow** (secondary tillage), and **Planter**. We also covered safety and maintenance (cleaning, oiling, sharpening).
- **Week 3 (Soil Fertility):** We defined fertility (soil's ability to supply nutrients).
 - **Loss:** Leaching, Erosion, Over-cropping.
 - **Improvement: Organic Manures** (compost, animal manure) and **Inorganic Fertilizers** (NPK, Urea).
 - **Compost Making:** Layering Greens (Nitrogen) and Browns (Carbon).
- **Week 4 (Land Prep 2):**
 - **Irrigation:** Adding water (drip, sprinkler).
 - **Drainage:** Removing excess water (ditches).

- **Crop Rotation:** Planting different crops in sequence to improve soil and control pests.
- **Week 6 (Site Selection):** Choosing a farm based on **Climatic** (rain, sun), **Soil** (type, fertility, topography), and **Economic** (market access, transport, labour) factors.
- **Week 8 (Planting):**
 - **Spacing:** Crucial to prevent *competition* for light, water, and nutrients.
 - **Direct Planting:** Seed in the main field (e.g., Maize).
 - **Transplanting:** Seedling from nursery to field (e.g., Tomato).
- **Week 9 (Weeds):** A weed is a plant in the wrong place.
 - **Effects:** Competition, alternate hosts for pests.
 - **Control:** Manual (hoeing, mulching), Chemical (herbicides), Biological (natural enemies).
- **Week 10 (Pests/Disease):**
 - **Pests:** Insects (borers, weevils), Vertebrates (rats, birds).
 - **Diseases:** Caused by pathogens (Fungi, Bacteria, Viruses).
 - **Control:** Physical (traps), Cultural (rotation), Chemical (pesticides), Biological (natural enemies).
 - **IPM:** A smart combination of all methods, using chemicals only as a last resort.

2. Class Activity: Rapid-Fire Q&A (Teacher asks quick-answer questions to review the entire term).

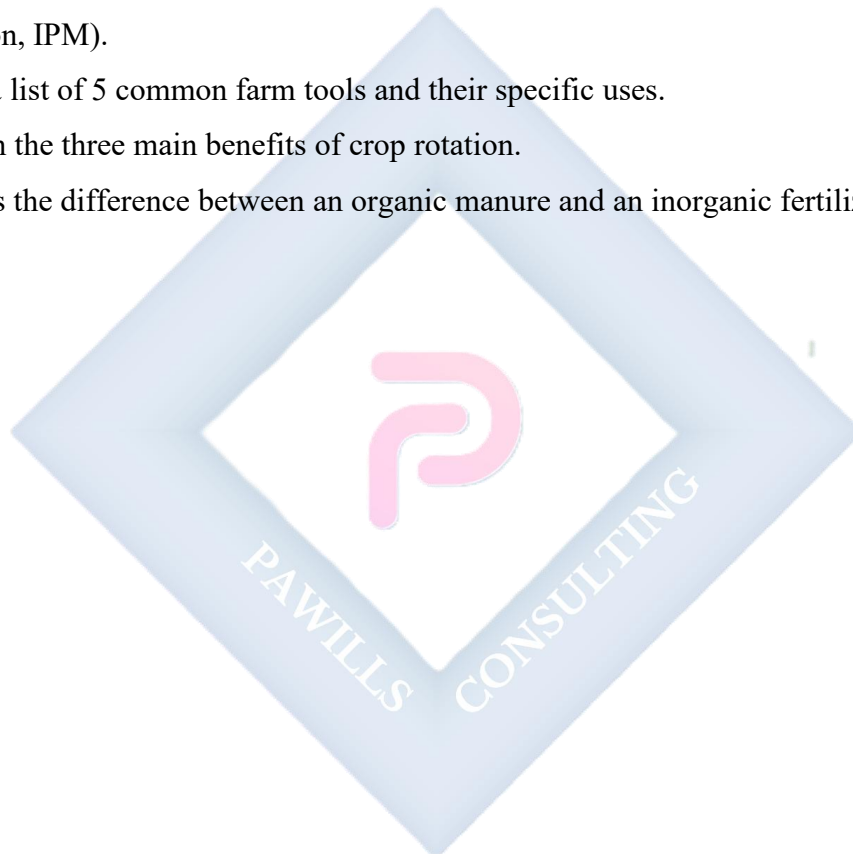
- "What tool is for primary tillage?" (Plough)
- "What tool is for secondary tillage?" (Harrow)
- "What is the 'washing down' of nutrients by rain called?" (Leaching)
- "What is the 'washing away' of topsoil called?" (Erosion)
- "NPK and Urea are what type of fertilizer?" (Inorganic/Chemical)
- "Dry leaves in a compost pile are 'Greens' or 'Browns'?" (Browns)
- "Applying water to a farm is called...?" (Irrigation)
- "Planting different crops in sequence is...?" (Crop Rotation)
- "Planting seeds in a nursery first is...?" (Transplanting)
- "A chemical that kills weeds is...?" (Herbicide)

- "A chemical that kills insects is..." (Insecticide)
- "What does IPM stand for?" (Integrated Pest Management)

Evaluation: This week's evaluation is the First Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key terms and their definitions (e.g., Tillage, Compost, Irrigation, Crop Rotation, IPM).
3. Make a list of 5 common farm tools and their specific uses.
4. Explain the three main benefits of crop rotation.
5. What is the difference between an organic manure and an inorganic fertilizer?



JSS 2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

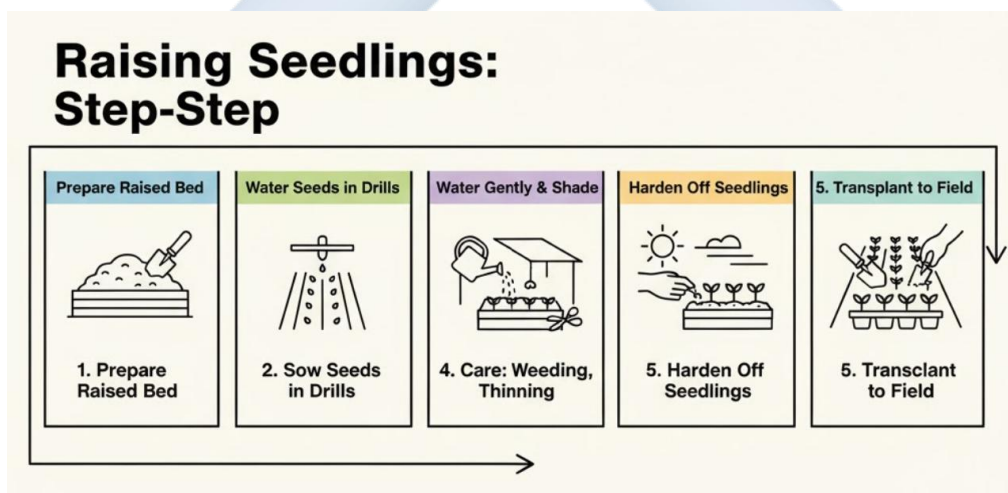
SECOND TERM

WEEK 1: NURSERY PRACTICES

Content: Preparation of nursery materials; Seed sowing and care; Transplanting techniques.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "nursery" and "nursery practices."
2. List the materials and tools needed for preparing a nursery bed.
3. Describe the steps involved in seed sowing in a nursery.
4. Explain the routine care given to seedlings in a nursery (e.g., watering, shading, weeding).
5. Define "transplanting" and "hardening off" and describe the transplanting process.



Lesson Content:

1. Review of Nursery

- **Definition:** A nursery is a special, well-prepared plot of land (or collection of containers) where young plants (seedlings) are raised with intensive care before they are moved to the permanent farm.
- **Nursery Practices:** These are all the activities involved in establishing and managing a nursery to ensure the production of healthy, strong seedlings.

2. Preparation of Nursery Materials and Site

- **Site Selection:** The site for a nursery must:
 - Be near a reliable water source.
 - Be on a flat, well-drained piece of land.
 - Be in a secure area, safe from pests and farm animals.
 - Have fertile, loamy soil.
- **Tools:** Hand trowel, hand fork, rake, watering can, measuring tape, pegs, and ropes.
- **Materials:**
 - **Seeds:** High-quality, viable seeds of the chosen crop (e.g., tomato, pepper).
 - **Soil:** Rich, loamy soil, often mixed with well-decomposed manure (compost).
 - **Shade:** Palm fronds, grass, or shade nets to protect seedlings from harsh sun.
 - **Water:** Clean water.
- **Bed Preparation:**
 1. Clear the chosen site of all weeds and stones.
 2. Till the soil (using a hoe, hand fork) to a fine tilth (soft, crumbly texture).
 3. Mark out the bed (e.g., 1m wide and any desired length).
 4. Create a raised bed (about 15-20cm high) by raking the tilled soil. This ensures good drainage.
 5. Level the top of the bed with a rake.

3. Seed Sowing and Care

- **Seed Sowing:**
 1. **Drills:** Use a small stick to create shallow, straight rows (drills) on the nursery bed, about 10-15cm apart.
 2. **Sowing:** Sow the tiny seeds *thinly* along the drills. Do not overcrowd them.
 3. **Covering:** Cover the seeds *lightly* with a thin layer of fine soil.
- **Post-Sowing Care (Routine Management):**

1. **Mulching:** Cover the bed with a thin layer of dry grass or leaves. This keeps the soil moist, prevents erosion, and stops the seeds from being splashed out. (Remove the mulch *immediately* after germination).
2. **Watering:** Water the bed gently *every day* (morning and evening) using a watering can with a "rose" head to prevent splashing.
3. **Shading:** Construct a *shade* (canopy) over the bed using palm fronds. This protects the tender seedlings from direct, hot sun and heavy rain.
4. **Weeding:** Remove any weeds *by hand* as soon as they appear. Weeds compete with the seedlings.
5. **Thinning:** If seedlings are overcrowded, pull out the weaker ones to give the strong ones more space to grow.
6. **Hardening Off:** This is the process of gradually *acclimatizing* the seedlings to the harsh conditions of the permanent farm.
 - **How:** About 1-2 weeks before transplanting, gradually *reduce* the amount of water and *remove the shade* for longer periods each day. This makes the seedlings tough.

4. Transplanting Techniques

- **Definition:** Transplanting is the careful process of moving a seedling from the nursery bed to its permanent position on the farm (ridge or field).
- **Procedure:**
 1. **Timing:** Transplanting is best done in the *late evening* or on a *cool, cloudy day* to reduce water loss (transpiration) and shock.
 2. **Watering:** Water the nursery bed thoroughly about an hour before transplanting to make the soil soft and easy to dig.
 3. **Uprooting:** Use a hand trowel to dig *around* and *under* a seedling. Lift it out carefully with a ball of earth *around its roots*. Never pull a seedling by its stem.
 4. **Planting:** Plant the seedling in the prepared hole on the permanent ridge. The hole should be deep enough to cover the ball of earth.
 5. **Firming:** Press the soil *gently* but *firmly* around the base of the seedling to remove air pockets and ensure root-soil contact.

6. **Watering:** Water the newly transplanted seedling immediately.

Evaluation:

1. What is a nursery?
2. List three materials needed for a nursery.
3. What is "shading" in a nursery, and why is it important?
4. What is "hardening off"?
5. Why is it best to transplant in the evening?

Assignment:

1. Define "nursery practices" and "transplanting."
2. Describe the process of "seed sowing" in a nursery.
3. List four routine care activities for seedlings in a nursery.
4. Why must you *never* pull a seedling by its stem when transplanting?
5. (Practical) If possible, construct a small nursery bed in your school or at home, sow some tomato seeds, and apply the post-sowing care.

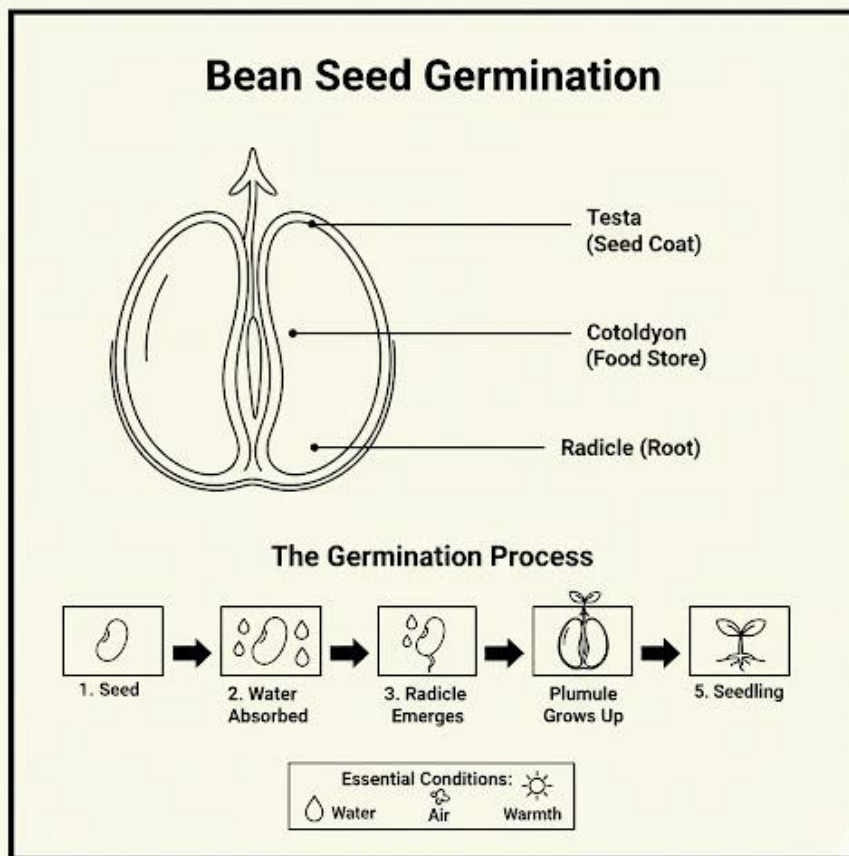
WEEK 2: SEED GERMINATION

Topic: Seed Germination

Content: Meaning and process; Conditions necessary for germination; Experiment on germination.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "seed" and "germination."
2. Draw and label the parts of a seed (e.g., a bean seed).
3. List the three *essential* conditions necessary for germination.
4. Describe the process of germination, step by step.
5. Set up a simple experiment to prove the conditions for germination.



Lesson Content:

1. Meaning and Process

- **Seed:** A seed is a small embryonic plant enclosed in a protective outer covering called a seed coat, usually with some stored food. It is the unit of reproduction for flowering plants.
- **Germination:** This is the *process* by which a dormant (sleeping) seed begins to grow and develop into a young plant (seedling) when all the right conditions are met.

2. Parts of a Seed (Example: Bean Seed)

- **Testa (Seed Coat):** The tough, protective outer layer.
- **Micropyle:** A tiny pore (hole) in the seed coat through which water enters.
- **Cotyledon(s):** The "seed leaves." They store food (starch, protein) for the embryo. (Beans have two, maize has one).
- **Embryo:** The "baby plant" itself, which consists of:
 - **Plumule:** Develops into the *shoot* (stem and leaves).
 - **Radicle:** Develops into the *root*.

3. Conditions Necessary for Germination For a viable (living) seed to germinate, it *must* have three external conditions:

1. Water (Moisture):

- Water enters through the micropyle and softens the testa.
- It "wakes up" the dormant embryo and activates the enzymes.
- It dissolves the stored food in the cotyledons so the embryo can use it.

2. Oxygen (Air):

- Once "awake," the embryo begins to respire (breathe).
- Respiration is the process of "burning" the stored food (like a tiny engine) to produce *energy* for growth.
- Food + Oxygen → Energy + Carbon Dioxide + Water

- If a seed is buried too deep or is in waterlogged soil, it gets no oxygen and will "drown" or rot.

3. Warmth (Suitable Temperature):

- Germination is a chemical process. All chemical reactions need a suitable temperature to work.
- Most crops (like maize, beans) prefer a *warm* temperature.
- If it is too cold or too hot, the seed will not germinate.

(Note: Light is not needed for germination. In fact, most seeds germinate best in the dark, underground.)

4. The Process of Germination (in a Bean Seed)

1. The seed absorbs **water** through the micropyle.
2. The testa softens and swells.
3. The embryo is activated and uses **oxygen** to respire.
4. The radicle (root) grows *downwards* into the soil to anchor the seedling and absorb more water.
5. The plumule (shoot) grows *upwards*, pulling the cotyledons with it, towards the light.
6. Once in the light, the cotyledons open up, and the first true leaves appear.
7. The seedling now starts *photosynthesis* (making its own food), and the cotyledons shrivel and fall off as their food store is used up.

5. Experiment: To Show the Conditions Necessary for Germination

- **Aim:** To prove that seeds need Water, Oxygen, and Warmth.
- **Apparatus:** 4 glass jars/cups, cotton wool, viable seeds (e.g., beans, maize).
- **Procedure:** Set up 4 jars as follows:
 - **Jar A (Control):** Contains *damp* cotton wool + seeds. Kept in a *warm* place (room temperature). (Has Water, Oxygen, Warmth).
 - **Jar B (No Water):** Contains *dry* cotton wool + seeds. Kept in a *warm* place. (Has Oxygen, Warmth; No Water).

- **Jar C (No Oxygen):** Contains *damp* cotton wool + seeds. Pour *boiled, cooled* water on top and cover with a layer of oil (to stop air from dissolving). Kept in a *warm* place. (Has Water, Warmth; No Oxygen).
- **Jar D (No Warmth):** Contains *damp* cotton wool + seeds. Kept in a *very cold* place (e.g., a refrigerator). (Has Water, Oxygen; No Warmth).
- **Observation:** After 3-4 days, you will observe...
- **Result:**
 - Jar A: Seeds germinate.
 - Jar B, C, D: Seeds **do not** germinate.
- **Conclusion:** This proves that all three conditions (Water, Oxygen, and Warmth) are *all* necessary for germination.

Evaluation:

1. What is "germination"?
2. What is the function of the cotyledon?
3. What is the function of the radicle?
4. List the three *essential* conditions for germination.
5. Why is oxygen needed for germination?

Assignment:

1. Draw a bean seed and label its four main parts (Testa, Micropyle, Cotyledon, Embryo).
2. Why is light *not* necessary for germination?
3. Describe the entire process of germination, from water absorption to the seedling stage.
4. In the experiment, why was the water in Jar C boiled? (To remove dissolved oxygen).
5. (Practical) Set up the germination experiment (Jar A and Jar B) at home and write down your observations after 3 days.

WEEK 3: FERTILIZER APPLICATION METHODS

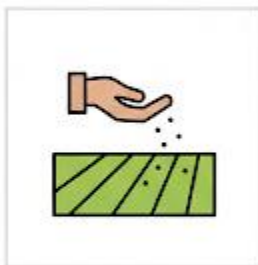
Topic: Fertilizer Application Methods

Content: Organic and inorganic fertilizers; Application methods and timing; Safety precautions.

Performance Objectives: By the end of this lesson, students should be able to:

1. Review the difference between organic and inorganic fertilizers.
2. List and describe three methods of fertilizer application.
3. Define "broadcasting," "ring application," and "foliar application."
4. State the importance of *timing* in fertilizer application.
5. List three safety precautions to take when handling chemical fertilizers.

NUTRIENT APPLICATION METHODS



BROADCASTING



RING METHOD



FOLIAR

Lesson Content:

1. Review of Fertilizer Types (from Term 1)

- **Organic Fertilizers (Manures):**
 - **Source:** Natural (animal waste, decomposed plants - compost).
 - **Action:** Slow-release of nutrients.
 - **Benefit:** Improves soil structure, water-holding capacity, and nutrients.
 - **Examples:** Poultry manure, cow dung, compost.
- **Inorganic Fertilizers (Chemical):**
 - **Source:** Man-made, factory-produced chemicals.
 - **Action:** Fast-release of *specific* nutrients.
 - **Benefit:** Provides a quick "boost" to plants. Does *not* improve soil structure.
 - **Examples:** NPK 15:15:15, Urea (provides Nitrogen).

2. Timing of Fertilizer Application

- **When to Apply:** Fertilizer is *not* applied during planting. The seed has its own food (cotyledon).
- **First Application (Top-Dressing):** Applied to the soil *around* the young crop, usually 2-4 weeks *after* germination (e.g., for maize, tomato). This gives the young plant a boost for vegetative (leafy) growth.
- **Second Application:** Often applied just *before* the flowering or fruiting stage to provide energy for producing fruits/seeds.
- **Key Rule:** Apply fertilizer when the soil is *moist* (e.g., after rain) so the nutrients can dissolve and be absorbed. *Never* apply to dry, hard soil.

3. Methods of Fertilizer Application

- **1. Broadcasting:**
 - **Definition:** The *scattering* of fertilizer (usually granular) evenly over the *entire surface* of the farm.

- **When to Use:** Best for crops that are planted very close together (e.g., rice, wheat, lawns) or for applying manure *before* ploughing.
- **Disadvantage:** Can be wasteful, as much of it falls where there are no roots.
- **2. Ring Application (or Spot Application):**
 - **Definition:** Applying fertilizer in a *circular band (ring)* around the *base* of an individual plant.
 - **Procedure:**
 1. Make a small, shallow trench (ring) about 5-10cm *away* from the plant's stem.
 2. Sprinkle the correct dose of fertilizer into the ring.
 3. Cover the fertilizer *lightly* with soil.
 - **Why?** Placing it *away* from the stem prevents the "burning" (scorching) of the plant. Covering it prevents the sun from destroying the nutrients (especially Nitrogen).
 - **When to Use:** Best for row crops like maize, tomato, pepper, and fruit trees.
- **3. Foliar Application:**
 - **Definition:** Mixing a *liquid* fertilizer with water and *spraying* it directly onto the *leaves* of the plant.
 - **Action:** The leaves absorb the nutrients (especially micronutrients) very quickly.
 - **When to Use:** This is used as a *quick-fix* to correct a nutrient deficiency, not as the main feeding method.
 - **Tool:** Knapsack sprayer.
- **4. Band Application:** Placing the fertilizer in a "band" or "row" *beside* the row of seeds during planting (done by a mechanical planter).

4. Safety Precautions (for Chemical Fertilizers)

1. **Read the Label:** Always read and follow the manufacturer's instructions for dosage and safety.
2. **Wear PPE:** Wear gloves, boots, and a simple nose/mouth mask to avoid skin contact and inhaling the dust.
3. **No Eating/Smoking:** Never eat, drink, or smoke while handling fertilizers.

4. **Wash Hands:** Wash your hands and face thoroughly with soap and water *after* application.
5. **Wind Direction:** Apply in the direction of the wind (so it blows *away* from you, not *into* your face).
6. **Storage:** Store fertilizers in a cool, dry, locked shed, away from food, water, and children.

Evaluation:

1. What is the main difference in the *action* of organic vs. inorganic fertilizers?
2. What is "broadcasting" of fertilizer?
3. Describe the "ring application" method.
4. Why must fertilizer be placed *away* from the plant stem?
5. List three safety precautions for handling chemical fertilizers.

Assignment:

1. Define Broadcasting, Ring Application, and Foliar Application.
2. When is the best *time* to apply fertilizer to a young maize plant?
3. What does "PPE" stand for, and give two examples used when applying fertilizer?
4. What is the main disadvantage of the "broadcasting" method?
5. (Practical) Demonstrate "ring application" on a plant in the school garden (you can use sand or sawdust to represent the fertilizer).

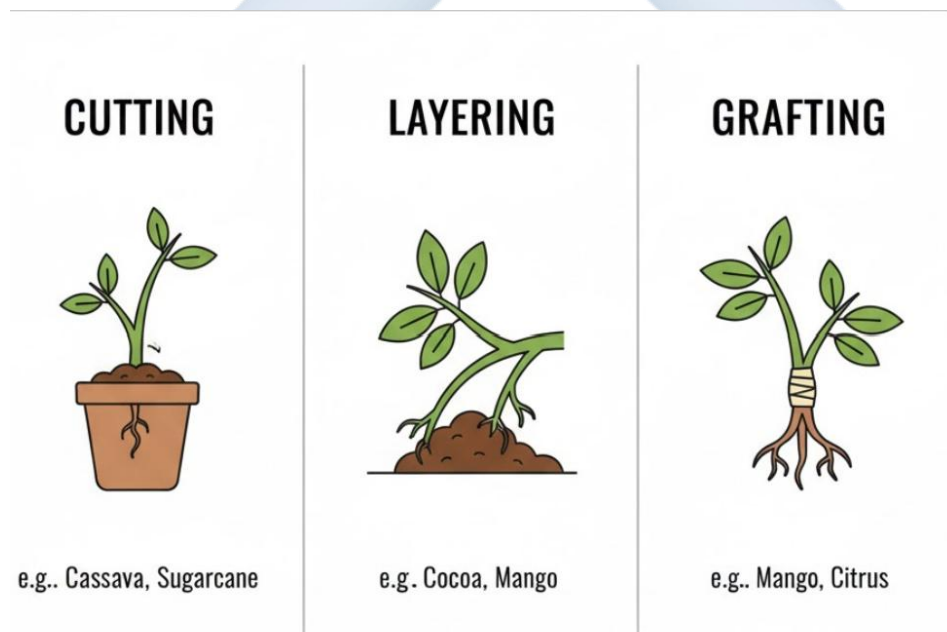
WEEK 4: VEGETATIVE PROPAGATION

Topic: Vegetative Propagation

Content: Meaning and importance; Methods of propagation: grafting, layering, cutting; Crops that use each method.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "vegetative propagation."
2. Differentiate between sexual propagation (seeds) and asexual (vegetative) propagation.
3. List three main advantages of vegetative propagation.
4. Define and describe "cutting" and give one crop example.
5. Define and describe "grafting" and "layering" and give crop examples.



Lesson Content:

1. Meaning of Vegetative Propagation

- **Sexual Propagation:** This is the *normal* way plants reproduce, using **seeds**. This involves the fusion of male and female gametes, so the offspring is a *mix* of two parents.

- **Asexual (Vegetative) Propagation:** This is the *artificial* method of growing a new plant from a *vegetative part* (like a stem, root, or leaf) of a *single parent plant*.
- The new plant is an *exact copy* (a "clone") of the parent plant.

2. Importance (Advantages) of Vegetative Propagation

1. **Genetic Uniformity:** The new plant is an *exact clone* of the parent. This means a sweet orange tree will produce *only* sweet oranges. (With seeds, you might get a sour orange).
2. **Faster Maturity:** Plants grown from cuttings or grafts will produce fruit *much faster* (e.g., 2-3 years) than plants grown from seed (e.g., 7-10 years).
3. **Propagation of Seedless Fruits:** This is the *only* way to grow fruits that do not have seeds (e.g., some types of oranges, grapes, bananas).
4. **Pest/Disease Resistance:** You can combine a strong plant with a tasty plant (see Grafting).

3. Methods of Vegetative Propagation

- **A. Cutting:**
 - **Definition:** This is the *simplest* method. A piece of the parent plant's *stem*, *root*, or *leaf* is cut off and planted in moist soil. It develops its own roots and grows into a new plant.
 - **Example (Stem Cutting):**
 1. Cut a 15-20cm piece of healthy stem from the parent plant (e.g., Cassava).
 2. Plant it directly in the field, often at an angle.
 - **Crops:** Cassava (most common), Sugarcane, Yam (cuttings of the tuber), Pineapple (the crown), Sweet Potato (vines).
- **B. Layering:**
 - **Definition:** This is the method of *forcing* a stem to grow roots *while it is still attached to the parent plant*.
 - **Methods:**

- **Simple Layering:** A low-growing, flexible branch is bent down, *pegged* into a small hole in the soil, and *covered* with soil (leaving the tip exposed).
 - **Air Layering (Marcotting):** Used for branches that are too high to bend. A small ring of bark is *removed* from the branch. This "wound" is covered with a ball of *moist soil* or moss and wrapped tightly with a plastic sheet.
 - **Result:** Roots will grow from the "wounded" part of the stem. Once the new roots are strong, the branch is cut off from the parent and planted.
 - **Crops:** Citrus (Orange, Lemon), Mango, Guava, Cocoa.
- **C. Grafting:**
 - **Definition:** This is the advanced technique of *joining* the parts of *two different plants* so they grow as one single plant.
 - **Parts:**
 - **Stock (Rootstock):** The *bottom* part (the rooted stem). It is chosen for its *strong roots* and *disease resistance*.
 - **Scion:** The *top* part (a small bud or shoot). It is chosen for its *high-quality fruit* (e.g., sweet taste).
 - **Procedure:** A "T-budding" or "Whip-graft" cut is made on the stock. The scion is inserted into the cut and tied tightly.
 - **Result:** The scion (sweet fruit) grows on the roots of the stock (strong plant).
 - **Crops:** Mango, Citrus (Orange, Lemon), Avocado, Apple, Cashew.

Evaluation:

1. What is "vegetative propagation"?
2. What is the main difference between sexual (seed) and vegetative (asexual) propagation?
3. List two *advantages* of using vegetative propagation.
4. What is "cutting"? Name one crop propagated by cuttings.
5. What is "grafting"? What are the "stock" and the "scion"?

Assignment:

1. Define Cutting, Layering, and Grafting.
2. Give one example of a crop for each method (Cutting, Layering, Grafting).
3. Why would a farmer choose to use "grafting" instead of just planting a seed? (Give 2 reasons).
4. What is "air layering"?
5. (Practical) Try to propagate a plant at home or school using the "cutting" method (e.g., with a cassava, hibiscus, or sweet potato stem).



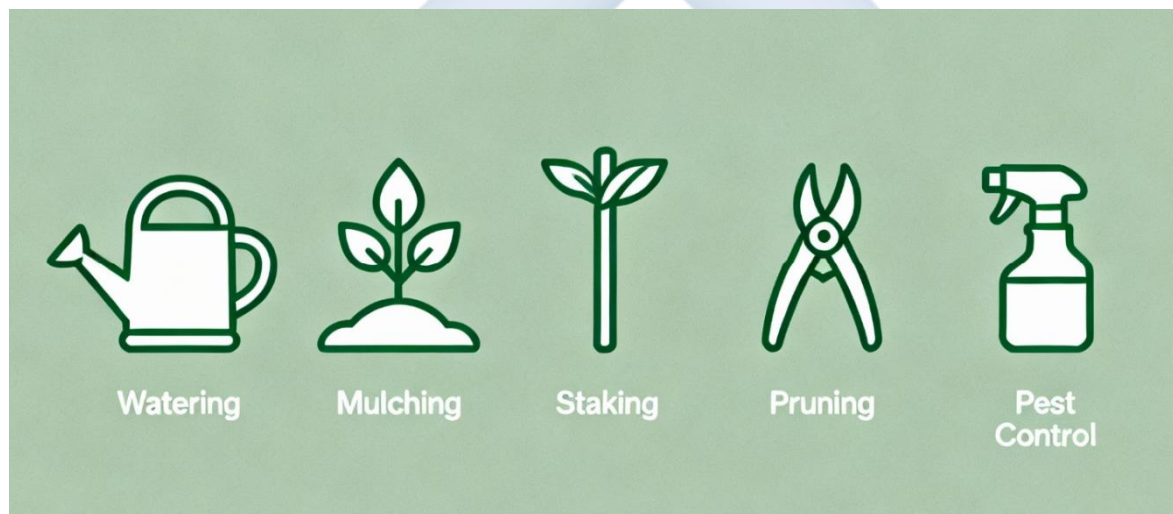
WEEK 6: MAINTENANCE OF HORTICULTURAL CROPS

Topic: Maintenance of Horticultural Crops

Content: Pruning, staking, watering, and mulching; Pest and disease control.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "maintenance" in crop production.
2. Define and state the importance of "pruning."
3. Define and state the importance of "staking."
4. Define and state the importance of "mulching."
5. Review the methods of pest and disease control in horticulture.



Lesson Content:

1. What are Crop Maintenance Practices?

- **Definition:** These are the regular *post-planting* activities (after planting or transplanting) that a farmer does to care for the crops, ensuring they grow healthily and produce a high yield.

2. Key Maintenance Practices

- **A. Watering (Irrigation):**
 - **Definition:** Supplying water to crops, especially during the dry season or in nurseries.
 - **Method:** Drip irrigation, sprinklers, or manual watering cans.
 - **Importance:** Water is essential for photosynthesis and for transporting nutrients.
- **B. Mulching:**
 - **Definition:** The process of *covering* the soil surface around the base of a plant with a layer of organic or inorganic material.
 - **Materials:**
 - *Organic:* Dry grass, straw, wood chips, leaves.
 - *Inorganic:* Black plastic (polythene) sheets.
 - **Importance (Benefits):**
 1. **Conserves Moisture:** Reduces evaporation of water from the soil surface.
 2. **Controls Weeds:** Blocks sunlight, preventing weed seeds from germinating.
 3. **Regulates Soil Temperature:** Keeps the soil cooler in the day and warmer at night.
 4. **Adds Nutrients:** Organic mulch will decompose and add humus to the soil.
- **C. Staking:**
 - **Definition:** The practice of providing a *support* (a stake, pole, or trellis) for weak-stemmed or "creeping" (vine) plants to climb on.
 - **Materials:** Bamboo poles, strong sticks, wooden trellis, ropes.
 - **Crops that need staking:** **Yam** (very important), Tomato (especially tall varieties), climbing Beans, Cucumber.
 - **Importance:**
 0. **Prevents Lodging:** Stops the plants from falling over due to wind, rain, or heavy fruit.
 1. **Better Sunlight Exposure:** Lifts the leaves up to get maximum sunlight for photosynthesis.

2. **Reduces Disease:** Keeps the leaves and fruits *off* the damp soil, which reduces rotting and fungal diseases.
 3. **Makes Harvesting Easier:** The fruits are easy to see and pick.
- **D. Pruning (or Trimming):**
 - **Definition:** The *selective removal* of certain parts of a plant, such as unwanted branches, leaves, or buds.
 - **Tools:** Secateurs (pruning shears), hand saw, or a sharp cutlass.
 - **Crops:** Fruit trees (Mango, Cocoa, Citrus), Tomato, Roses.
 - **Importance:**
 0. **Removes Diseased/Dead Parts:** Stops the spread of disease.
 1. **Improves Airflow:** Opens up the plant's canopy, which reduces disease.
 2. **Controls Growth:** Gives the plant a good, manageable shape.
 3. **Increases Fruit Size:** By removing *some* excess flowers or small fruits, the plant's energy is directed to the *remaining* fruits, making them larger and sweeter.
 - **E. Pest and Disease Control:**
 - **Review (from Term 1):**
 - **Pests:** Insects, rats, birds.
 - **Diseases:** Fungi, bacteria, viruses.
 - **Control:**
 - **Cultural:** Crop rotation, sanitation (removing sick plants).
 - **Physical:** Hand-picking pests, traps, fences.
 - **Chemical:** Spraying *insecticides* (for insects) or *fungicides* (for fungi).
 - **IPM:** Using a smart combination of all methods.

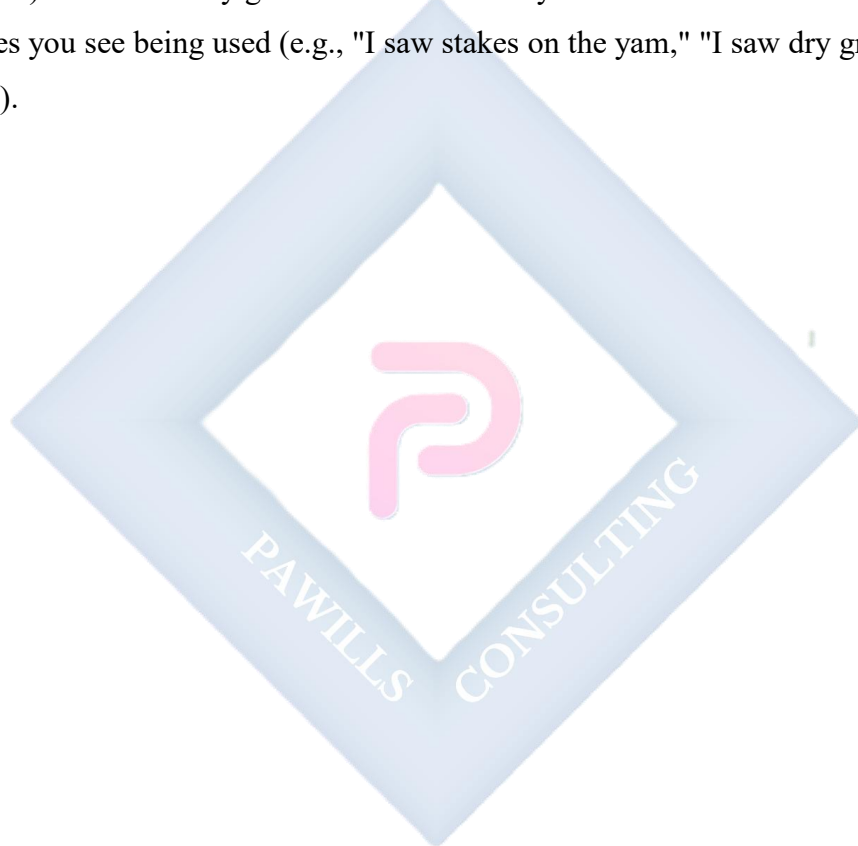
Evaluation:

1. What are "crop maintenance practices"?
2. What is "staking," and name one crop that needs it.
3. What is "mulching"?
4. List two *benefits* of mulching.

5. What is "pruning," and state one reason why it is done.

Assignment:

1. Define Staking, Mulching, and Pruning.
2. Give one example of an *organic* mulching material and one *inorganic* material.
3. List three *benefits* of staking your yam or tomato plants.
4. Why would a farmer *prune* (remove) some of the flowers from a tomato plant?
5. (Practical) Go to a nearby garden or farm. Identify and write down *three* maintenance practices you see being used (e.g., "I saw stakes on the yam," "I saw dry grass used as mulch").



WEEK 8: HARVESTING AND STORAGE

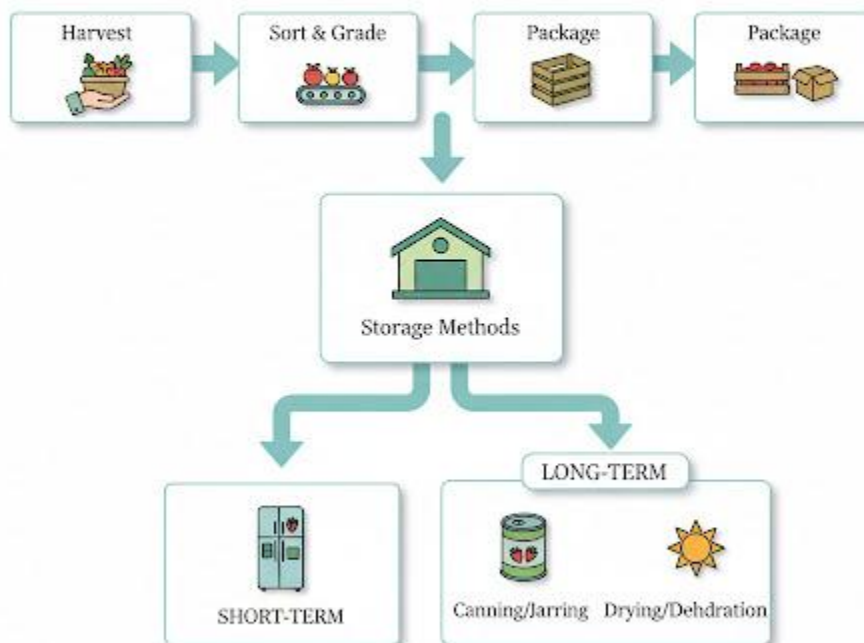
Topic: Harvesting and Storage

Content: Techniques of harvesting; Post-harvest handling and preservation.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "harvesting," "maturity," and "post-harvest."
2. Describe two different harvesting techniques (e.g., digging, picking).
3. List three problems of "post-harvest" handling.
4. Define "storage" and "preservation."
5. Describe two methods of crop preservation (e.g., drying, refrigeration).

Post-Harvest Produce Flow



Lesson Content:

1. Harvesting

- **Definition:** Harvesting is the process of *gathering* or *collecting* mature crops from the farm.
- **Maturity:** This is the stage when the crop is *ready* to be harvested.
 - **Physiological Maturity:** The plant has completed its growth (e.g., grain is hard, fruit is sweet).
 - **Harvest Maturity:** The specific time the crop is harvested to meet market demands (e.g., some bananas are harvested *green* for shipping).
- **Timing:** Harvesting must be done at the *right time*.
 - **Harvesting too early (immature):** Crop quality is low, it won't store well.
 - **Harvesting too late (over-mature):** Crop may rot, get eaten by pests, or spoil.
- **Harvesting Techniques (Methods):**
 - **Manual (by hand):**
 - **Picking:** For fruits and vegetables (e.g., tomato, pepper, okra, oranges).
 - **Digging (Lifting):** For root/tuber crops (e.g., yam, cassava, potato) using a hoe or spade.
 - **Cutting:** For cereals or vegetables (e.g., rice, sugarcane, cabbage) using a sickle or cutlass.
 - **Mechanical (by machine):**
 - Using **Harvesters** (e.g., Combine Harvester for maize/rice, Tomato Harvester).

2. Post-Harvest Handling

- **Definition:** This includes *all* the activities that happen *immediately after* harvesting, before the crop is stored or sold.
- **Process:**
 1. **Gathering:** Bringing the crops from the farm to one central point.

2. **Cleaning:** Washing (e.g., carrots) or "threshing" (e.g., beating beans to remove pods) to remove dirt, stones, and unwanted parts.
 3. **Sorting & Grading:** Separating the crops based on *quality, size, or ripeness* (e.g., Grade 1 tomatoes are big and red; Grade 2 are small).
 4. **Packaging:** Putting the crops into bags, crates, or baskets for easy transport.
- **Post-Harvest Losses:** A large amount of food (30-50%) is lost in Nigeria *after* harvesting.
 - **Causes of Loss:**
 - **Mechanical Injury:** Bruising, cutting, or breaking the crop during careless harvesting and transport.
 - **Pest/Disease:** Pests (weevils) and fungi (mould) attacking the harvested crop.
 - **Environmental:** Rotting or sprouting due to high heat and humidity.

3. Storage and Preservation

- **Storage:** The process of *keeping* farm produce safe for future use, in a way that maintains its quality.
- **Preservation:** The specific *treatment* or *process* used to *stop* food from spoiling (rotting, decaying).
- **Methods of Preservation (Traditional and Modern):**
 1. **Drying (Dehydration):**
 - **Method:** Removing water from the food. (Micro-organisms cannot grow without water).
 - **Examples:** Sun-drying of maize, beans, pepper, okra; smoking of fish/meat.
 2. **Refrigeration / Freezing (Cold Storage):**
 - **Method:** Using *low temperatures* to slow down the growth of bacteria and fungi.
 - **Examples:** Storing fresh tomatoes, fruits, and vegetables in a refrigerator or cold room.
 3. **Canning / Bottling:**

- **Method:** Sealing the food in an airtight container (can or bottle) and then heating it to kill all micro-organisms.
 - **Examples:** Canned tomatoes, canned sweet corn, bottled fruit juices.
4. **Use of Chemicals:**
- **Method:** Adding *preservatives* like salt, sugar, or vinegar.
 - **Examples:** *Salting* fish; *sugaring* to make fruit jams; *pickling* with vinegar.
5. **Barns and Silos:**
- **Method:** Large, specialized structures for storing grain (silos) or tubers (yam barns). They are designed to be cool, dry, and rodent-proof.

Evaluation:

1. What is "harvesting"?
2. What does "maturity" mean?
3. List two common harvesting techniques (e.g., picking, digging).
4. What is "post-harvest loss"?
5. What is the *main principle* behind "drying" as a preservation method?

Assignment:

1. Define Harvesting, Storage, and Preservation.
2. What is "grading," and why is it important?
3. List three common causes of post-harvest losses.
4. Describe two traditional methods of preservation and two modern methods.
5. (Practical) Look in your kitchen at home. Find and list three food items that are *preserved* and state the method used (e.g., "Tin Tomatoes - Canning," "Dry Fish - Drying/Smoking").

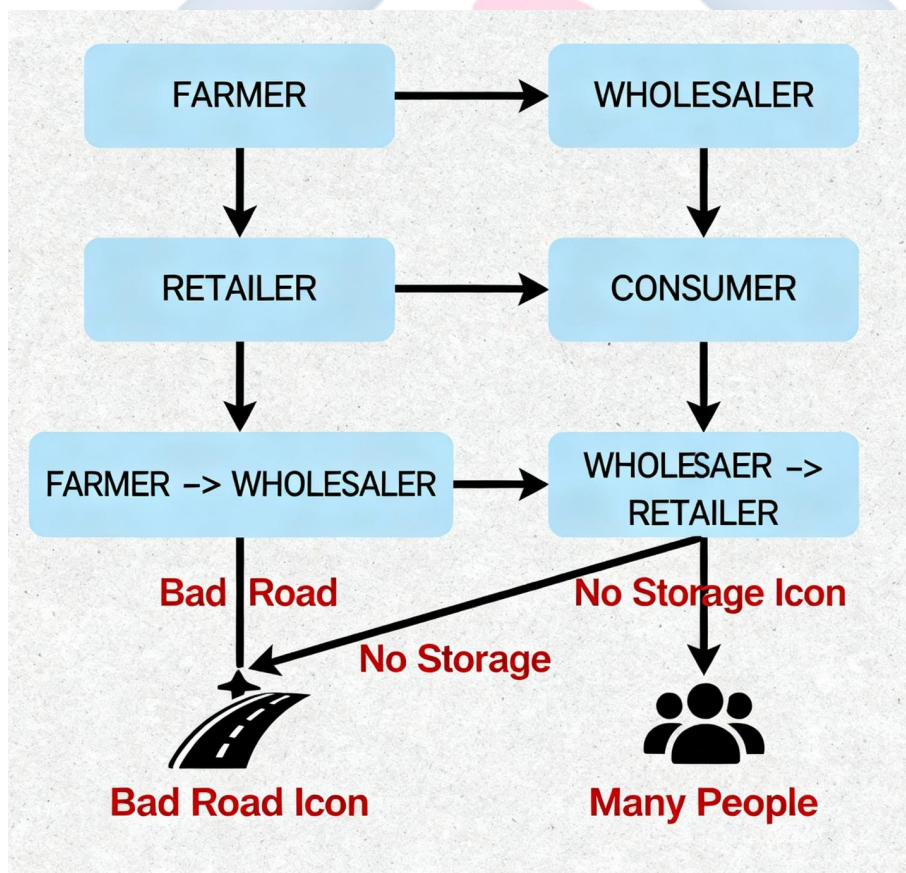
WEEK 9: MARKETING OF HORTICULTURAL CROPS

Topic: Marketing of Horticultural Crops

Content: Meaning and importance of marketing; Channels of distribution; Marketing problems and solutions.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "marketing" in agriculture.
2. State three reasons why marketing is important.
3. Define "channel of distribution."
4. Draw a simple diagram showing a channel of distribution (e.g., Farmer -> Wholesaler -> Retailer -> Consumer).
5. List three major problems of agricultural marketing in Nigeria.



Lesson Content:

1. Meaning and Importance of Marketing

- **Definition:** Agricultural Marketing includes *all* the business activities involved in moving a farm product (like tomatoes) from the *farm* to the *final consumer*.
- It is *not* just "selling." It includes:
 - **Harvesting**
 - **Grading and Sorting**
 - **Packaging**
 - **Transportation**
 - **Storage**
 - **Processing** (e.g., turning oranges into juice)
 - **Advertising and Sales**
- **Importance of Marketing:**
 1. **Provides Income:** It is the *only* way a farmer gets paid for their hard work.
 2. **Prevents Spoilage:** An efficient marketing system (with good storage and transport) moves perishable crops (like tomatoes) to the consumer *before* they spoil.
 3. **Links Producer and Consumer:** It bridges the gap between the farmer in the village and the consumer in the city.
 4. **Creates Jobs:** Marketing provides jobs for drivers, packagers, wholesalers, retailers, and advertisers.

2. Channels of Distribution

- **Definition:** A channel of distribution is the *path* or *route* that a product takes to get from the producer (farmer) to the final consumer.
- **Key "Middlemen" (Intermediaries):**
 - **Wholesaler (Distributor):** Buys in *very large quantities* from many farmers, stores the produce, and sells in *large quantities* to retailers.

- **Retailer:** Buys in *large quantities* from the wholesaler and sells in *small quantities* (e.g., 1kg, 5 tomatoes) directly to the final consumer. (e.g., The local market woman).
- **Common Channels:**
 - **Channel 1 (Shortest):**
 - Producer (Farmer) → Final Consumer
 - (e.g., A farmer at a roadside stall).
 - **Channel 2 (Most Common):**
 - Producer (Farmer) → Retailer → Final Consumer
 - **Channel 3 (Longest):**
 - Producer (Farmer) → Wholesaler → Retailer → Final Consumer

3. Problems of Agricultural Marketing in Nigeria

1. **Poor Transportation:** Bad roads damage crops during transport, increase costs, and cause delays, leading to spoilage.
2. **Lack of Storage Facilities:** Farmers have no cold rooms or proper silos. This forces them to sell *immediately* at harvest time when prices are *low*. This leads to "seasonal scarcity" (food is cheap at harvest, very expensive later).
3. **Poor Packaging:** Using rough baskets and raffia crushes and bruises crops (especially tomatoes).
4. **Too Many Middlemen:** In Channel 3, every middleman adds their own profit. This makes the food *very expensive* for the final consumer, but the *farmer* still gets paid very little.
5. **Lack of Market Information:** Many farmers in villages don't know the real price of their crops in the city, so middlemen cheat them.

4. Solutions to Marketing Problems

1. **Government:** Repairing rural roads.

2. **Cooperatives:** Farmers should form **cooperatives** (groups) to buy their own storage (silos) and transport (trucks), so they can "cut out" the middlemen and sell directly to retailers, keeping more profit.
3. **Technology:** Using mobile phones to check market prices before selling.
4. **Improved Packaging:** Using plastic or wooden crates for transport.

Evaluation:

1. Define "agricultural marketing."
2. Is marketing just "selling"? Explain.
3. Who is a "wholesaler"?
4. Who is a "retailer"?
5. List two major problems of agricultural marketing in Nigeria.

Assignment:

1. Define Marketing and Channel of Distribution.
2. Draw a diagram showing the 3-step channel of distribution (Farmer to Consumer).
3. What is a "farmers' cooperative"?
4. How can poor roads and lack of storage cause "post-harvest loss"?
5. Explain how "too many middlemen" is a problem for *both* the farmer and the final consumer.

WEEK 10: FARM RECORD-KEEPING

Topic: Farm Record-Keeping

Content: Types and importance of records; Cashbook, sales record, stock record.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "farm records."
2. State four reasons why keeping records is important.
3. List and define four different types of farm records.
4. Differentiate between a "sales record" and an "expenditure record."
5. Draft a simple stock record for a farm.



Lesson Content:

1. What are Farm Records?

- **Definition:** Farm records are a detailed, written account of all activities, events, and transactions that happen on a farm.
- **It is the "memory" and "diary" of the farm.** A farmer's brain cannot remember everything; the book does.

2. Importance of Farm Records

1. **To Calculate Profit/Loss:** This is the *most important* reason. By comparing income (sales) and expenses, the farmer knows if the farm is a *business* or a *hobby*.
2. **To Secure Loans:** A bank will *not* give a loan to a farmer who has no records. Records *prove* that the farm is a real, profitable business.
3. **To Improve Management:** Records help the farmer make *smart decisions*.
 - (e.g., "My records show that my poultry feed costs are too high. I need to find a cheaper source.")
 - (e.g., "My records show that the 'B' variety of maize produced a higher yield than the 'A' variety. Next year, I will only plant 'B'.")
4. **For Tax Assessment:** Used to calculate how much tax the farm needs to pay.
5. **For Insurance:** If there is a disaster (like a fire), the records are used to claim insurance.

3. Types of Farm Records

- **1. Farm Diary / Daily Log:**
 - **Purpose:** To record all *daily activities* on the farm.
 - **Content:** "Monday: Weeded the tomato plot." "Tuesday: Applied NPK fertilizer to the maize." "Wednesday: 10 new chicks hatched."
- **2. Production / Yield Record:**
 - **Purpose:** To track *how much* the farm is producing.
 - **Content:** "Date: 10/10/25. Crop: Maize. Yield: 15 bags." "Date: 11/10/25. Crop: Eggs. Yield: 5 crates."

- **3. Stock / Inventory Record:**

- **Purpose:** To track all the *items* (stock) on the farm.
- **Content:** Lists all tools, machines, feeds, fertilizers, and animals. It shows "Quantity In," "Quantity Out," and "Balance" (what is left).
- *Example (Feed Store):* | Date | Item | Quantity In (Bags) | Quantity Out (Bags) | Balance (Bags) || :--- | :--- | :--- | :--- | :--- || Oct 1 | Starter Mash | 20 (Bought) || 20 || Oct 2 | Starter Mash | 2 (Used) | 18 || Oct 3 | Starter Mash | 2 (Used) | 16 |

- **4. Financial Records (The "Money Books"):**

- **A. Expenditure (Expense) Record:** Records all money *spent* (e.g., buying seeds, feed, paying labour).
- **B. Sales (Income) Record:** Records all money *received* from sales (e.g., selling eggs, yams).
- **C. Cashbook:** A "master" book that combines *all* income and *all* expenses to find the final profit or loss.

Evaluation:

1. What are "farm records"?
2. State the *most important* reason for keeping records.
3. How do records help a farmer get a bank loan?
4. What is the difference between an "Expenditure Record" and a "Sales Record"?
5. What would you write in a "Farm Diary"?

Assignment:

1. Define "Farm Records."
2. List three *types* of farm records.
3. List three *reasons* why farm records are important.
4. (Practical) A farmer buys 50 bags of fertilizer on Monday and uses 10 bags on Tuesday. Draw a simple **Stock Record** to show this.
5. (Practical) The farmer sells 3 goats for N15,000 each. Which record book does this go into? (Sales Record).

Week 11: REVISION

Topic: Revision

Content: Review of all topics in Term 2

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term (e.g., Nursery, Germination, Propagation, Staking, Mulching, Marketing, Records).
2. Explain the difference between direct planting and transplanting.
3. Explain the difference between sexual (seed) and vegetative (asexual) propagation.
4. Describe the three main methods of vegetative propagation.
5. Explain the importance of marketing and farm records.

Lesson Content:

1. Summary of Term 2 Topics

- **Week 1 (Nursery):** We learned about "nursery practices": preparing a bed, sowing seeds in drills, and post-sowing care (watering, shading, weeding, thinning). We learned "transplanting" (moving seedlings to the field) and "hardening off" (making them tough).
- **Week 2 (Germination):** We defined germination (a seed sprouting). We learned the 3 conditions: **Water, Oxygen, Warmth**. We learned the *process*: the *radicle* (root) grows down, and the *plumule* (shoot) grows up.
- **Week 3 (Fertilizer):** We reviewed organic vs. inorganic. We learned application *methods*: **Broadcasting** (scattering), **Ring Application** (in a circle, away from the stem), and **Foliar** (spraying on leaves). We also learned safety rules (wear PPE, wash hands).
- **Week 4 (Vegetative Propagation):** This is asexual reproduction (cloning) from a parent plant.
 - **Advantages:** Faster, identical (clones), good for seedless fruits.
 - **Methods:**
 - **Cutting:** Planting a piece of stem (e.g., Cassava, Sweet Potato).

- **Layering:** Growing roots on a branch *before* cutting it (e.g., Citrus, Mango).
- **Grafting:** Joining a **Scion** (good fruit) to a **Stock** (strong root) (e.g., Mango, Orange).
- **Week 6 (Maintenance):** These are "post-planting" activities:
 - **Mulching:** Covering the soil (with grass/plastic) to conserve water and stop weeds.
 - **Staking:** Using a pole to support weak-stemmed plants (e.g., Yam, Tomato).
 - **Pruning:** Cutting off unwanted branches to improve airflow and fruit size.
- **Week 8 (Harvest/Storage):**
 - **Harvesting:** Gathering mature crops.
 - **Post-Harvest:** Cleaning, grading, and packaging.
 - **Preservation:** Stopping food from spoiling (e.g., **Drying, Refrigeration, Canning, Salting**).
- **Week 9 (Marketing):** All activities moving food from farm to consumer.
 - **Channel:** Producer → Wholesaler → Retailer → Consumer.
 - **Problems:** Bad roads, no storage, too many middlemen.
- **Week 10 (Records):** Writing down all farm activities.
 - **Importance:** To calculate profit, get loans, and make smart decisions.
 - **Types:** Farm Diary, Stock Record, Financial Records (Sales, Expenses).

2. Class Activity: Rapid-Fire Q&A

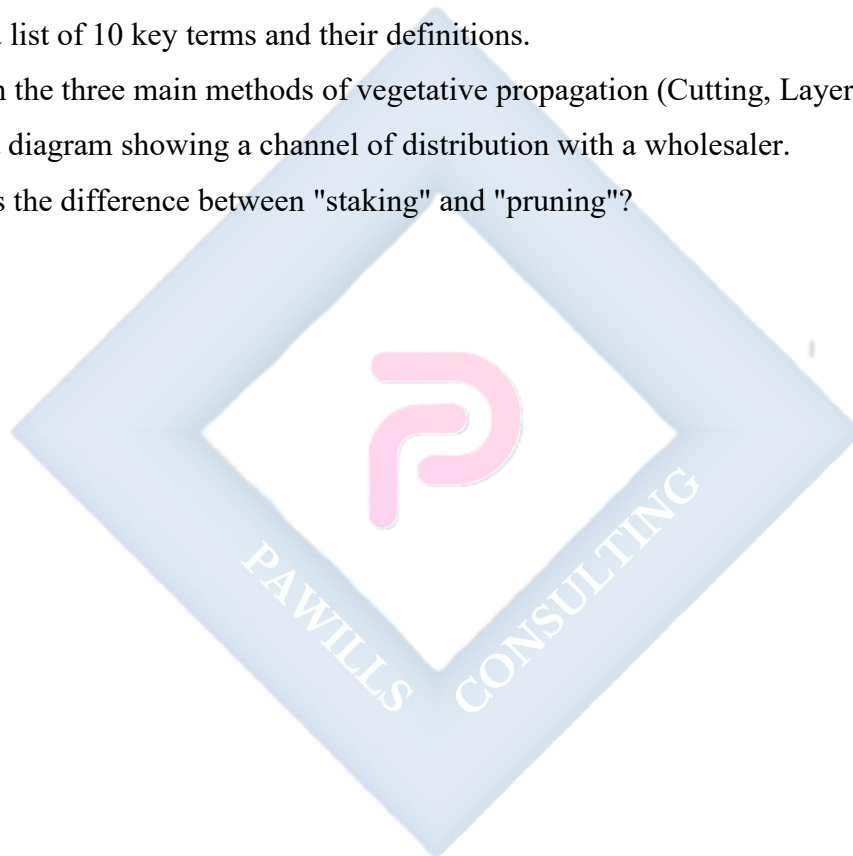
- "What do we call the baby root of an embryo?" (Radicle)
- "What do we call the baby shoot?" (Plumule)
- "What are the 3 conditions for germination?" (Water, Oxygen, Warmth)
- "A chemical fertilizer like NPK is..." (Inorganic)
- "Growing a new plant from a cassava stem is..." (Cutting)
- "Joining two plants (stock and scion) is..." (Grafting)
- "Using a pole to support a tomato plant is..." (Staking)
- "Covering the soil with dry grass is..." (Mulching)
- "A chemical that kills insects is..." (Insecticide)

- "Removing water from maize is...?" (Drying)
- "The person who buys from the farmer and sells to the market woman is...?" (Wholesaler)
- "A book for tracking all money spent is...?" (Expenditure Record)

Evaluation: This week's evaluation is the Second Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key terms and their definitions.
3. Explain the three main methods of vegetative propagation (Cutting, Layering, Grafting).
4. Draw a diagram showing a channel of distribution with a wholesaler.
5. What is the difference between "staking" and "pruning"?



JSS 2 HORTICULTURE AND CROP PRODUCTION LESSON NOTES

THIRD TERM

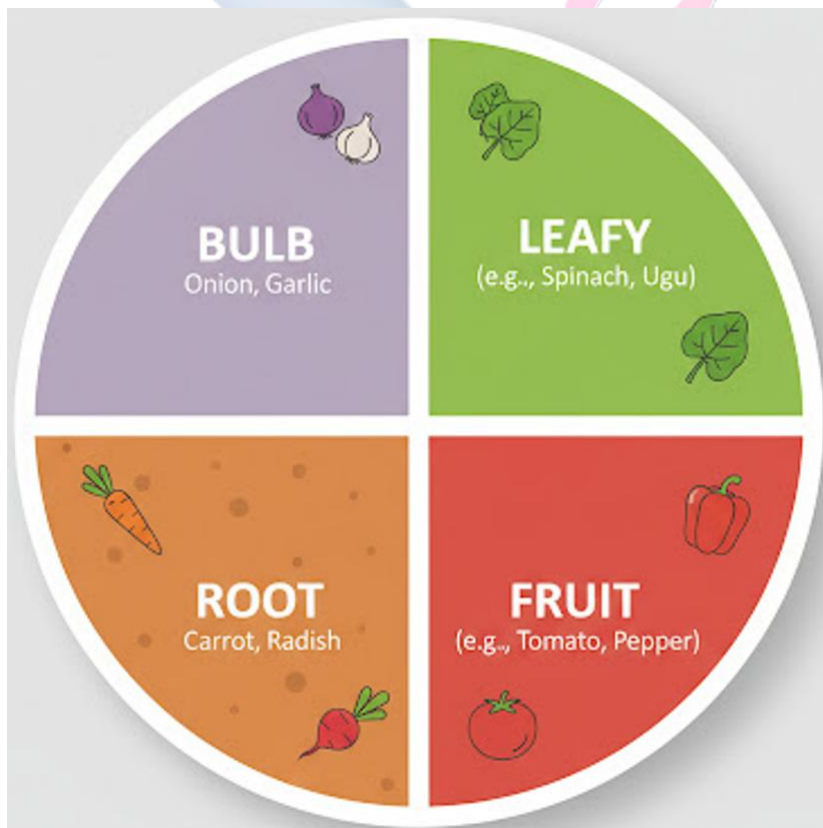
WEEK 1: VEGETABLE PRODUCTION

Topic: Vegetable Production

Content: Importance and types of vegetables; Conditions for growth; Production stages.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "vegetable" and state its nutritional and economic importance.
2. Classify vegetables into at least three groups, giving two examples for each.
3. List the major conditions necessary for successful vegetable growth.
4. Describe the key stages involved in the production of a common vegetable (e.g., tomato or spinach).



Lesson Content:

1. Importance and Types of Vegetables

- **Definition:** Vegetables are the edible parts of plants, such as the leaves, stems, roots, tubers, or fruits, that are commonly consumed by humans. Horticulture is the branch of agriculture that deals with the intensive cultivation of vegetables, fruits, and flowers.
- **Importance of Vegetables:**
 - **Nutritional Importance:** They are a primary source of essential **vitamins** (e.g., Vitamin A for sight, Vitamin C for skin/immunity) and **minerals** (e.g., Iron for blood, Calcium for bones). They also provide **dietary fibre** (roughage), which is essential for healthy digestion.
 - **Economic Importance:** Vegetable farming is a major source of **income** for many families. It creates **employment** (on the farm, in transport, at the market) and contributes to the nation's **food security**.
- **Classification (Types) of Vegetables:**
 1. **Leafy Vegetables:** These are plants whose *leaves* are eaten.
 - *Examples:* Spinach (*Efo*, *Amunututu*), Fluted Pumpkin (*Ugu*), Waterleaf, Lettuce, Cabbage.
 2. **Fruit Vegetables:** These are plants whose *fruits* are eaten as vegetables (biologically they are fruits, but in the kitchen, they are used as vegetables).
 - *Examples:* Tomato, Pepper (all types), Okra, Eggplant (Garden Egg), Cucumber.
 3. **Root/Tuber Vegetables:** These are plants whose edible part grows *underground*.
 - *Examples:* Carrot, Radish, Turnip, Beetroot.
 4. **Bulb Vegetables:** These are vegetables with a swollen underground bulb.
 - *Examples:* Onion, Garlic, Spring Onion.

2. Conditions for Vegetable Growth Vegetables are generally high-value but delicate crops. They require specific conditions to thrive:

1. **Soil:** They need deep, fertile, well-drained **loamy soil** that is rich in organic matter (compost or manure). Heavy clay soil or pure sandy soil is not ideal.
2. **Climate (Temperature & Sunlight):** Most vegetables require warm temperatures and *full, direct sunlight* (at least 6-8 hours a day) for photosynthesis.
3. **Water:** Vegetables have shallow roots and need a *consistent, regular, and reliable* supply of water. They cannot tolerate drought. Access to irrigation (drip, sprinkler, or manual) is essential.
4. **Site:** The land must be flat or gently sloping (to prevent erosion) and located in a secure area (to prevent theft or damage by animals).

3. Production Stages (Example: Tomato Production)

1. **Site Selection:** Choose a sunny, well-drained site with fertile soil, near a water source.
2. **Land Preparation:** Clear the land, till the soil (plough and harrow) to a fine tilth, and make raised beds or ridges. Mix well-rotted manure or compost into the soil.
3. **Propagation (Planting):**
 - Tomatoes are first planted in a **nursery** (See Term 2, Week 1).
 - Seeds are sown in drills in the nursery, watered, and shaded.
 - Seedlings are "hardened off" after 3-4 weeks.
 - **Transplanting:** Seedlings are transplanted to the permanent beds in the evening, at a spacing of about 60cm x 60cm.
4. **Maintenance Practices:**
 - **Watering:** Water every 1-2 days if there is no rain.
 - **Weeding:** Remove weeds regularly, either by hand or with a hoe.
 - **Fertilizer Application:** Apply NPK fertilizer using the "ring method" (See Term 2, Week 3) about 2-3 weeks after transplanting.
 - **Staking:** Provide a 1-2m stake (stick) for each plant to climb on. This keeps the fruits off the ground, prevents disease, and makes harvesting easy (See Term 2, Week 6).
 - **Pruning:** (Optional) Removing some of the side shoots ("suckers") to encourage the plant to produce fewer, but larger, fruits.

- **Pest/Disease Control:** Watch for insects (like fruit borers) and diseases (like blight). Spray with appropriate (safe) insecticides or fungicides if necessary.

5. Harvesting:

- Tomatoes are ready for harvest about 2-3 months after transplanting.
- They are harvested by *hand-picking* the mature fruits (which can be green-mature or fully red, depending on the market).
- Harvesting is done every 3-5 days for several weeks.

Evaluation:

1. Define "vegetables" and state two reasons why they are important.
2. Classify the following: (a) Okra, (b) Ugu, (c) Carrot.
3. List three essential conditions for growing vegetables.
4. Why are most vegetables transplanted from a nursery?
5. What is "staking," and why is it important for tomato production?

Assignment:

1. Define "Horticulture."
2. List the three main classes of vegetables and give two examples for each.
3. What is the ideal soil type for vegetable production?
4. Describe, step-by-step, the "production stages" for a leafy vegetable like Spinach (*Efo*). (Hint: It is planted *directly*, not transplanted).
5. (Practical) Start a small vegetable garden at home or in school (e.g., in a bucket, used tyre, or small bed). Plant some Ugu, waterleaf, or okra seeds and care for them.

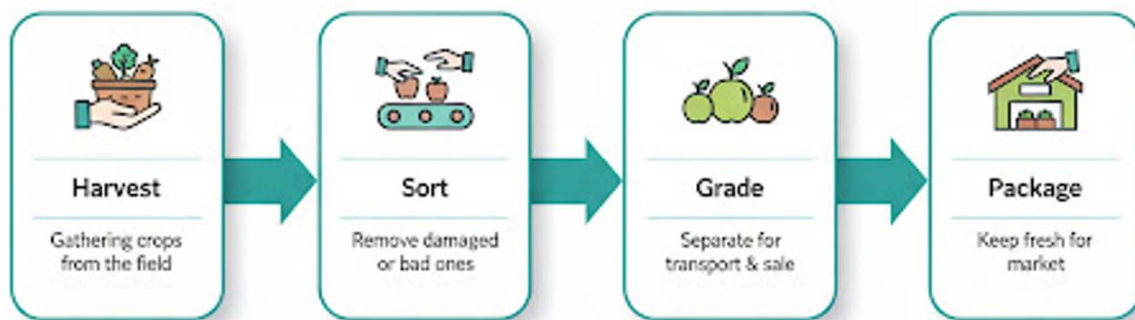
WEEK 2: POST-HARVEST MANAGEMENT

Topic: Post-Harvest Management

Content: Sorting, grading, and packaging; Storage methods.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "post-harvest management."
2. Differentiate between "sorting" and "grading."
3. State the criteria used for grading horticultural crops.
4. Describe three different packaging materials and their uses.
5. Explain two short-term and two long-term storage methods for vegetables.



Lesson Content:

1. Review of Post-Harvest (from Term 2)

- **Definition:** Post-harvest management (or "post-harvest handling") refers to *all* the activities performed on a crop *immediately after* it is harvested from the farm, to protect its quality, prevent spoilage, and prepare it for the market or storage.
- **Importance:** Horticultural crops (vegetables and fruits) are *highly perishable* (they spoil quickly). Good post-harvest management is essential to reduce "post-harvest losses" and increase the farmer's profit.

2. Sorting, Grading, and Packaging

- **A. Sorting:**

- **Definition:** This is the process of *removing* unwanted materials from the harvested produce.
- **Process:** It involves separating the *good* produce from the *bad* produce.
- **Items Removed:**
 - Diseased, rotten, or pest-damaged produce.
 - Immature or over-ripe produce.
 - Damaged or bruised produce (from harvesting).
 - Foreign materials (e.g., dirt, stones, leaves, weeds).

- **B. Grading:**

- **Definition:** This is the *next step* after sorting. It involves categorizing the *good* produce into different "grades" or classes based on *quality*.
- **Criteria for Grading:**
 1. **Size:** e.g., Grade 1 (Large), Grade 2 (Medium), Grade 3 (Small).
 2. **Colour:** e.g., Fully red tomatoes vs. partially red tomatoes.
 3. **Firmness/Texture:** e.g., Hard, firm fruits vs. slightly soft ones.
 4. **Freedom from Blemishes:** e.g., Perfect skin vs. skin with small spots.
 - **Importance:** Grading *increases the value* of the produce. A farmer can sell "Grade 1" tomatoes at a much higher price to hotels, while selling "Grade 2" at a lower price in the open market.

- **C. Packaging:**

- **Definition:** The process of enclosing the graded produce in a container for protection, handling, transport, and marketing.
- **Importance:** Good packaging *protects* the produce from being crushed, bruised, or contaminated.
- **Packaging Materials:**
 - **Traditional (Bad):** Raffia baskets, woven bags. They are cheap but offer poor protection; fruits are easily crushed.

- **Modern (Good):**

- **Plastic Crates (Perforated):** Strong, stackable, and allow for air circulation (ventilation). Ideal for tomatoes, peppers.
- **Wooden Crates:** Strong, good for transporting yams or citrus.
- **Net Bags:** Used for onions, potatoes, cabbage (allows ventilation).
- **Cardboard Cartons:** Used for high-value or export crops.

3. Storage Methods

- **Definition:** Storage is the process of keeping farm produce in a safe and controlled environment to preserve its quality and prevent spoilage until it is needed.
- **A. Short-Term Storage (Days to Weeks):**
 1. **Cool, Dark, Ventilated Place:** Many vegetables (like yams, potatoes, onions) can be stored for a few weeks in a cool, dry, and dark room with good air movement.
 2. **Refrigeration (Cold Storage):** Using refrigerators or "cold rooms" (at 0-10°C). This is the *best* method for highly perishable leafy greens, tomatoes, and fruits. It slows down respiration and the growth of micro-organisms.
- **B. Long-Term Storage (Months to Years) (Processing/Preservation):**
 1. **Drying (Dehydration):** Removing water to stop microbes from growing.
 - *Examples:* Sun-drying of peppers, okra, or leafy vegetables (to make "Gbure").
 2. **Canning / Bottling:** Sealing the produce (e.g., tomatoes, sweetcorn) in an airtight can or jar and heating it to kill all bacteria.
 3. **Processing:** Changing the form of the produce.
 - *Examples:* Processing tomatoes into *paste/puree*; processing peppers into *dry chili powder*.

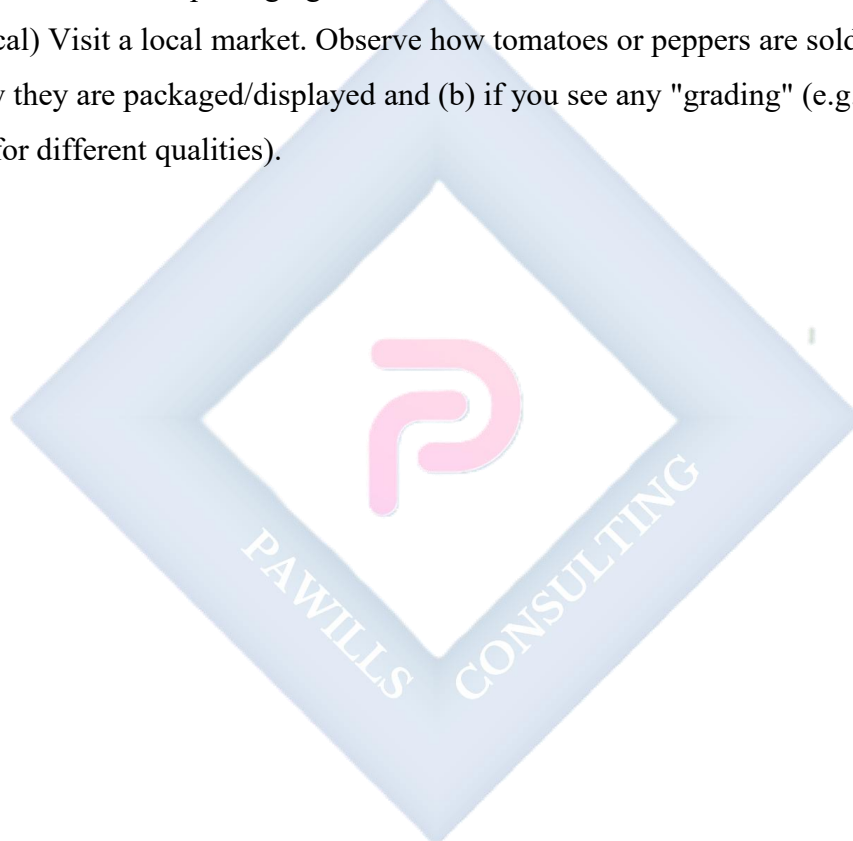
Evaluation:

1. What is "post-harvest management"?
2. What is the difference between "sorting" and "grading"?
3. List three criteria used for grading tomatoes.

4. Why is a plastic crate better for packaging tomatoes than a raffia basket?
5. What is the *best* short-term storage method for leafy vegetables?

Assignment:

1. Define Sorting, Grading, and Packaging.
2. List three common post-harvest activities.
3. Why is "grading" important for a farmer's income?
4. Describe two modern packaging materials and two traditional ones.
5. (Practical) Visit a local market. Observe how tomatoes or peppers are sold. Write down (a) how they are packaged/displayed and (b) if you see any "grading" (e.g., different prices for different qualities).



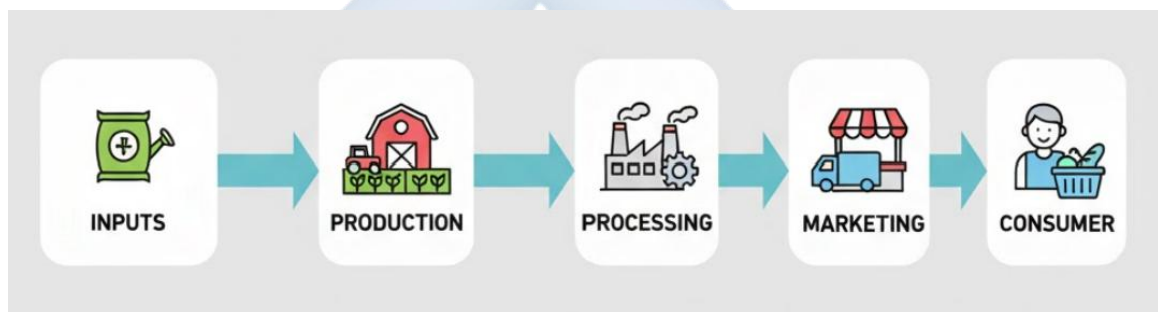
WEEK 3: INTRODUCTION TO AGRIBUSINESS

Topic: Introduction to Agribusiness

Content: Meaning of agribusiness; Career opportunities.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "agribusiness."
2. Explain the concept of the "agribusiness value chain."
3. List the four main components of the agribusiness sector.
4. Identify at least two career opportunities in each component.



Lesson Content:

1. Meaning of Agribusiness

- **Definition:** "Agribusiness" is the combination of the words "Agriculture" and "Business."
- It refers to *all* the business activities related to agriculture. It is not just *farming*; it includes every step, from the factory that makes the fertilizer to the supermarket that sells the food.
- Agribusiness sees farming as a modern, commercial, and profitable enterprise, not just a poor man's subsistence activity.

2. The Agribusiness Value Chain A "value chain" describes all the stages a product goes through, where "value" is added at each stage.

Farm → Processing → Distribution → Consumer

3. Components and Career Opportunities in Agribusiness The entire agribusiness sector can be broken down into four main components:

- **A. Input Supply Sector:**

- **Definition:** This sector provides all the "inputs" (materials) that the farmer *needs* to start production.
- **Career Opportunities:**
 1. **Seed and Fertilizer Sales:** Owning an "agro-dealer" shop.
 2. **Feed Mill Operator:** Producing and selling animal feed.
 3. **Tool/Machine Manufacturing:** Making or selling cutlasses, tractors, or irrigation equipment.
 4. **Agricultural Finance:** Working in a bank that gives loans to farmers.

- **B. Production Sector (The Farm):**

- **Definition:** This is the *actual farming* part—growing crops and raising animals.
- **Career Opportunities:**
 1. **Crop Farmer:** (e.g., a commercial vegetable farmer, a rice plantation owner).
 2. **Livestock Farmer:** (e.g., a poultry farmer, a fish farmer).
 3. **Horticulturist:** A specialist in growing flowers, fruits, and vegetables.
 4. **Farm Manager:** A professional hired to manage a large commercial farm.

- **C. Processing Sector (Post-Harvest):**

- **Definition:** This sector *transforms* raw farm produce into finished, more valuable products. This is also called "Value Addition."
- **Career Opportunities:**
 1. **Food Processor:** (e.g., turning cassava into garri, tomatoes into paste, oranges into juice, groundnuts into oil).
 2. **Packaging Specialist:** Designing attractive and safe packaging for food.
 3. **Butcher/Meat Processor:** Turning live animals into packaged meat, sausages, etc.

4. **Food Quality Control Officer:** Working in a factory to ensure food is safe.

- **D. Marketing and Support Sector:**

- **Definition:** This sector handles the *movement* of the product to the consumer and provides all other necessary services.
- **Career Opportunities:**
 1. **Transporter:** Owning trucks to transport food from farms to cities.
 2. **Marketer/Advertiser:** Promoting and selling the processed food.
 3. **Wholesaler/Retailer:** (Middlemen, market owners).
 4. **Support Services:**
 - **Agricultural Extension Agent:** A government or private worker who *teaches* farmers new techniques.
 - **Agricultural Researcher/Scientist:** Develops new, better seeds and methods.

Evaluation:

1. What is "agribusiness"?
2. Is agribusiness just "farming"? Explain your answer.
3. What are the four main components of the agribusiness sector?
4. Giving an example, what is "value addition" (processing)?
5. List one career in the "Input Supply" sector and one in the "Processing" sector.

Assignment:

1. Define "agribusiness" in your own words.
2. What is an "agribusiness value chain"?
3. List the four components of agribusiness and give one example of a business in each.
4. A person who owns a tractor and ploughs farms for other farmers for a fee is in which sector of agribusiness?
5. A person who designs labels for bottled orange juice is in which sector?



WEEK 4: ENTREPRENEURSHIP IN AGRICULTURE

Topic: Entrepreneurship in Agriculture

Content: Meaning and qualities of an entrepreneur; Importance of entrepreneurship.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "entrepreneur" and "entrepreneurship."
2. List at least five key qualities (characteristics) of a successful entrepreneur.
3. Define "agripreneur."
4. Explain three reasons why entrepreneurship is important for national development.



Lesson Content:

1. Meaning of Entrepreneur and Entrepreneurship

- **Entrepreneur:** An entrepreneur is a person who *identifies a business opportunity* (a problem to solve or a need to fill), gathers the resources (money, people, materials), and *starts a business* to provide a product or service, taking on *financial risks* in the hope of making a *profit*.
 - *Example:* A person who sees that people in their town have no access to fresh eggs, so they take a risk (their savings) to start a poultry farm.
- **Entrepreneurship:** This is the *process* or *act* of being an entrepreneur—the act of setting up and running a new business.
- **Agriprenuer:** A modern term for an **entrepreneur in agribusiness**.

2. Qualities (Characteristics) of a Good Entrepreneur What makes someone a successful entrepreneur?

1. **Risk-Taker:** They are willing to risk their time, money, and energy on an idea that might fail.
2. **Innovative/Creative:** They think of new ideas, new products, or new ways of doing things. (e.g., "Instead of just selling raw tomatoes, I will process them into chili sauce!").
3. **Persistent and Hard-working:** They do not give up easily when they face challenges (and they *will* face challenges). They are resilient and determined.
4. **Decisive and Proactive:** They make firm decisions and take action. They don't wait for things to happen; they *make* things happen.
5. **Good Manager:** They know how to manage resources wisely (money, time, people).
6. **Good Communicator:** They can explain their ideas clearly to customers, investors, and employees.
7. **Opportunity-Seizes (Problem-Solver):** They see problems as *opportunities*.
 - *Problem:* "My neighbours' pepper spoils before it gets to market."
 - *Opportunity:* "I will start a small business drying and grinding the pepper into powder!"

3. Importance of Entrepreneurship in Agriculture Why does a country *need* agripreneurs?

1. **Job Creation:** When an entrepreneur starts a new farm or food processing company, they *hire people* (farm managers, drivers, factory workers, sales staff). This reduces unemployment.
2. **Food Security:** Agripreneurs produce food on a large scale, ensuring that there is enough food for the nation's population.
3. **Development of Rural Areas:** Most agricultural businesses are in rural areas. When a factory or farm is set up, it brings roads, electricity, and development to that village.
4. **Increases Gross Domestic Product (GDP):** New businesses create new wealth, which makes the entire country richer.
5. **Reduces Post-Harvest Losses:** Entrepreneurs (processors) buy raw produce from farmers and turn it into products with a long shelf-life (e.g., fruit juice, yam flour), thus saving food that would have rotted.
6. **Introduction of New Technology:** Entrepreneurs are often the first to adopt new technologies (like drip irrigation or modern silos) to make farming more efficient.

Evaluation:

1. Who is an entrepreneur?
2. What is an "agripreneur"?
3. List three qualities of a good entrepreneur.
4. List three reasons why entrepreneurship is important for a country.
5. How does an entrepreneur see a "problem"? (As an opportunity).

Assignment:

1. Define "Entrepreneurship."
2. List and *explain* three qualities of an entrepreneur (e.g., "Risk-taker: This means they are not afraid to...").
3. List and *explain* three reasons why entrepreneurship is important.
4. (Research) Name one famous Nigerian entrepreneur (in any field) and state the business they created.

5. (Personal) Think of a simple *problem* in your school or neighbourhood. What *business opportunity* could solve it?



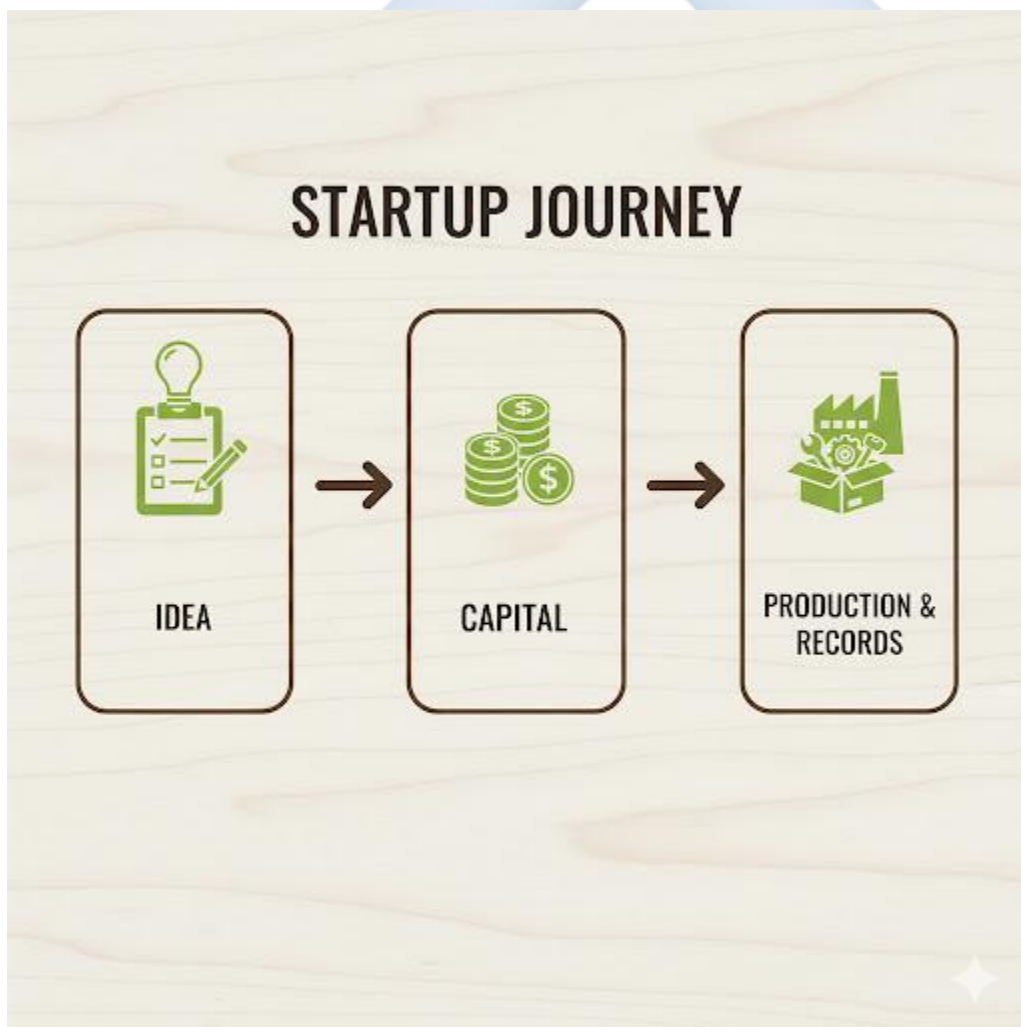
WEEK 6: ENTERPRISE DEVELOPMENT

Topic: Enterprise Development

Content: Small agribusiness setup; Record-keeping.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "enterprise" (a business).
2. List the first five basic steps in setting up a small agribusiness enterprise.
3. Define "business plan" and state its importance.
4. Review the key farm records needed for a new business (from Term 2, Week 10).
5. Draft a simple setup plan for a small enterprise (e.g., a small vegetable garden).



Lesson Content:

1. What is Enterprise Development?

- **Enterprise:** Another name for a business or a company.
- **Enterprise Development:** The *process* of starting and *growing* a new business. This lesson focuses on the *startup* phase for a small agribusiness.

2. Steps in Setting Up a Small Agribusiness Enterprise

1. Step 1: Idea Generation (The "What"):

- Identify an opportunity (a problem to solve).
- *Example Idea:* "People in my area want fresh vegetables, but the market is far."
- *Your Enterprise:* A small-scale vegetable farm (horticulture).

2. Step 2: Feasibility Study / Business Plan (The "How"):

- **Definition:** A **Business Plan** is a *written document* that describes your business, its goals, and how you will achieve them. It is your "roadmap to success."
- **Importance:** It forces you to *think* through all the details. It is also *required* to get a bank loan.
- **Simple Plan includes:**
 - **Goal:** What is my aim? (e.g., "To produce and sell 20kg of ugu and spinach weekly.")
 - **Market:** Who will I sell to? (e.g., "My neighbours and a local restaurant.")
 - **Requirements:** What do I need? (e.g., Land, seeds, 1 watering can, 1 hoe).
 - **Finances:** How much will it cost? (Capital).

3. Step 3: Source Capital (The "Money"):

- **Capital:** The money needed to *start* the business.
- **Sources for a small enterprise:**
 - Personal Savings (best)
 - Money from family and friends
 - Grants or loans from cooperatives or government programs.

4. Step 4: Acquire Resources / Setup (The "Tools"):

- Use the capital to *buy* the things in your business plan.
- *Example:* Secure the land (school garden), buy the seeds (ugu), buy the tools (hoe, watering can).

5. Step 5: Start Production (The "Work"):

- Begin the technical work (land preparation, planting, weeding, watering, etc.).

3. The Role of Record-Keeping in a New Enterprise (Review from Term 2, Week 10)

- From **Day 1**, you *must* keep records. A business with no records is a failed business.
- **Purpose:** To know if you are making a **profit** or a **loss**.
- **Key Records for a New Enterprise:**
 1. **Expenditure Record (Expenses):** Write down *everything* you spend money on.
 - *Example:* | Date | Item | Cost (N) | | :--- | :--- | :--- | | Nov 1 | Ugu Seeds (1 pack) | N300 | | Nov 1 | Watering Can (1) | N1,500 | | **Total** | | **N1,800** |
 2. **Production Record:** Write down *everything* you produce.
 - *Example:* | Date | Crop | Quantity Harvested | | :--- | :--- | :--- | | Dec 10 | Ugu | 50 bunches | | Dec 17 | Ugu | 60 bunches |
 3. **Sales Record (Income):** Write down *everything* you sell.
 - *Example:* | Date | Item Sold | Quantity | Price (N) | | :--- | :--- | :--- | :--- | | Dec 10 | Ugu | 50 bunches | N2,500 | | Dec 17 | Ugu | 60 bunches | N3,000 | | **Total** | | | **N5,500** |
- **Profit Calculation:** Total Sales (N5,500) - Total Expenses (N1,800) = **Profit (N3,700)**

Evaluation:

1. What is an "enterprise"?
2. What is a "business plan," and why is it important?
3. What is "capital"? List two places a JSS2 student could get capital.
4. What is the *most important* reason for keeping farm records?
5. What is the difference between an expenditure record and a sales record?

Assignment:

1. List the 5 main steps for setting up a small enterprise.
2. Define "Business Plan."
3. Define "Capital."
4. (Practical) Imagine you want to start a small *poultry* business with 10 chicks.
 - (a) List 3 things you would need (Requirements).
 - (b) List 3 things that would be in your "Expenditure Record."
 - (c) List 1 thing that would be in your "Sales Record."
5. What is the formula for calculating Profit?



WEEK 8: FINANCIAL MANAGEMENT

Topic: Financial Management

Content: Basic money management; Budgeting and planning.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "financial management."
2. Explain the importance of separating personal money from business money.
3. Define "budget" and "budgeting."
4. Explain the difference between "income" and "expenditure."
5. Draft a simple budget for a small farm project.



Lesson Content:

1. What is Financial Management?

- **Definition:** Financial management is the *process* of **planning, organizing, controlling, and monitoring** the financial resources (the money) of a business, in order to achieve its goals.
- **In simple terms:** It is "smart money management" for your business.

2. The First Rule: Separate Business and Personal Money

- This is the *most important rule* for a small entrepreneur.
- **Problem:** A farmer gets N5,000 from selling vegetables. He uses N1,000 for his child's school fees, N2,000 for food, and N500 for airtime. He only has N1,500 left. He *thinks* he is "making money," but he has no money to buy fertilizer for the next season. His business will die.
- **Solution:** The business must have its *own* money.
 1. Open a separate bank account for the business (or use a separate "savings box").
 2. All N5,000 from sales goes *into* the business account.
 3. The farmer pays himself a small "salary" (e.g., N1,000) from the business account *to* his personal pocket.
 4. The remaining N4,000 *stays* in the business account, ready to buy seeds, feed, and fertilizer for the next cycle. This is called **reinvestment**.

3. Budgeting and Planning

- **Income:** All money that comes *into* the business (e.g., sales).
- **Expenditure:** All money that goes *out* of the business (e.g., costs, expenses).
- **Budget:** A **budget** is a *plan* for your income and expenditure *before* you spend the money. It is a financial "roadmap" for a specific period (e.g., a month, a planting season).
- **Budgeting:** The *process* of creating this plan.

4. How to Create a Simple Farm Budget (Example: Budget for 1 poultry cycle - 6 weeks for broilers)

- **Goal:** To raise 20 broiler chicks.
- **1. Estimate Your Expenditures (Costs):**
 - *Capital Costs (One-time):*
 - Feeder (1): N1,000
 - Drinker (1): N800
 - *Operating Costs (Running):*
 - Day-old chicks (20 x N500): N10,000
 - Feed (Starter, Finisher): N15,000
 - Medication/Vaccines: N1,000
 - Wood shavings: N500
 - **TOTAL ESTIMATED EXPENDITURE = N28,300**
- **2. Estimate Your Income (Sales):**
 - *Goal:* Raise 18 chicks successfully (assume 2 die).
 - *Sale Price:* N3,500 per chicken (at 6 weeks).
 - **TOTAL ESTIMATED INCOME = 18 x N3,500 = N63,000**
- **3. Plan (Calculate Estimated Profit):**
 - Estimated Income: N63,000
 - Estimated Expenditure: (N28,300)
 - **ESTIMATED PROFIT = N34,700**
- **Why is this budget useful?**
 1. It tells the farmer they need **N28,300** *before* they can even start.
 2. It shows the *potential* profit, making it a good business.
 3. It acts as a *guide* for spending.

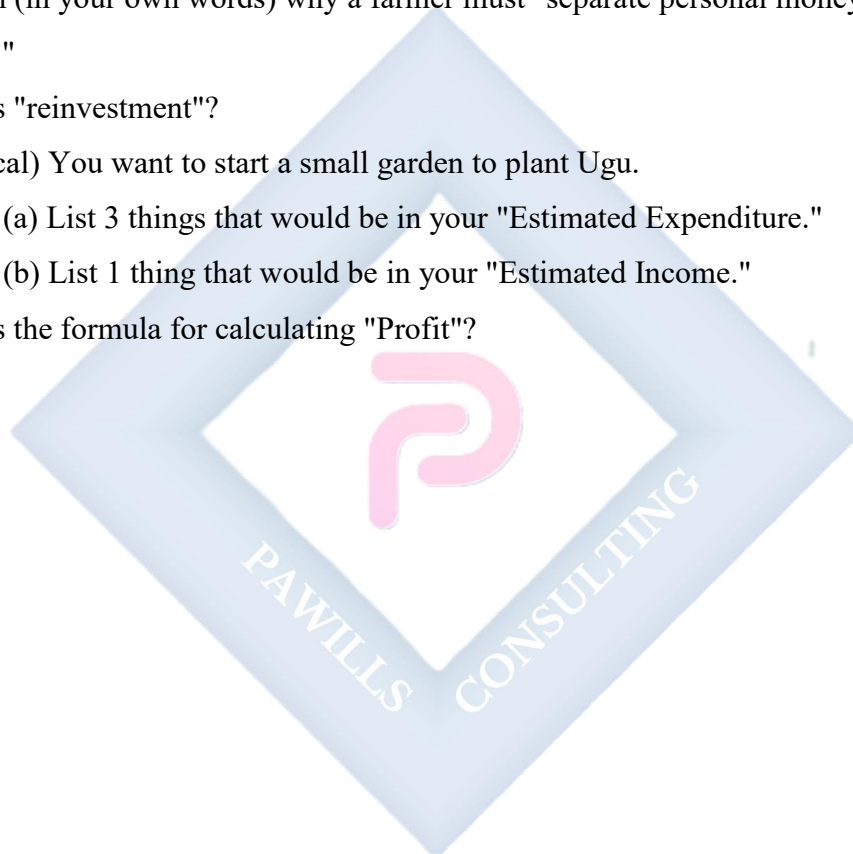
Evaluation:

1. What is "financial management"?

2. What is the *first and most important rule* of money management for a small business?
3. What is a "budget"?
4. What is the difference between "income" and "expenditure"?
5. Why is a budget useful? (Give one reason).

Assignment:

1. Define "Financial Management" and "Budget."
2. Explain (in your own words) why a farmer must "separate personal money from business money."
3. What is "reinvestment"?
4. (Practical) You want to start a small garden to plant Ugu.
 - (a) List 3 things that would be in your "Estimated Expenditure."
 - (b) List 1 thing that would be in your "Estimated Income."
5. What is the formula for calculating "Profit"?



WEEK 9: MARKETING AND PROMOTION

Topic: Marketing and Promotion

Content: Marketing strategies; Customer relations.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "marketing strategy."
2. List and explain the "4 Ps" of marketing (Product, Price, Place, Promotion).
3. Define "customer relations" and state its importance.
4. Describe three ways to maintain good customer relations.



Lesson Content:

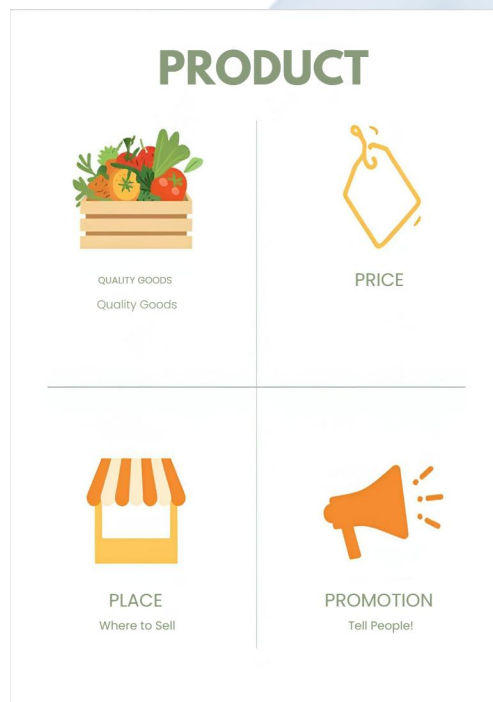
1. What is a Marketing Strategy?

- **Definition:** A marketing strategy is the business's *overall game plan* for how it will reach potential customers and turn them into actual customers.
- The most famous strategy is the "**4 Ps of Marketing.**"

2. The "4 Ps" of Marketing

• A. PRODUCT:

- This is *what* you are selling. It's not just the item; it's the *quality*.
- **Strategy:** Your product must be *good*.
- *Examples:* Are your vegetables fresh and clean? Are your eggs large and clean? Is your packaging (Week 2) attractive and protective? A good product sells itself.



• B. PRICE:

- This is *how much* you will sell the product for.
- **Strategy:** Your price must be *competitive* (not too high, not too low) and it *must* cover your costs and give you a **profit**.
- *Pricing Method:* (Cost of Production) + (Your Profit Margin) = **Selling Price**.

• C. PLACE:

- This is *where* your customers will find and buy your product.
- **Strategy:** You must sell *where your customers are*.
- *Examples:* Your farm gate, the local market, supplying directly to schools, restaurants, or hotels.

- **D. PROMOTION (Advertising):**

- This is *how* you *tell* people about your product.
- **Strategy:** You must communicate your value to the customer.
- *Examples for a small business:*
 - **Word of Mouth:** (Most powerful) Do a good job, and customers will tell their friends.
 - **Simple Sign/Banner:** A clear, well-painted sign at your farm gate or shop.
 - **Branding:** Give your farm a name (e.g., "Peaceful Farms") and put it on your packaging.
 - **Social Media:** Using WhatsApp Status, Facebook, or Instagram to show pictures of your fresh produce.

3. Customer Relations

- **Definition:** Customer relations (or Customer Service) is the way a business *communicates and interacts* with its customers.
- **Importance:** Good customer service is the *cheapest* and *best* form of marketing.
 - It is easier to *keep* an old customer than to find a new one.
 - A *happy* customer will come back and will tell 5 friends (Promotion).
 - An *unhappy* customer will *not* come back and will tell 20 friends (Bad Promotion).

4. How to Maintain Good Customer Relations

1. **Be Polite and Respectful:** Always greet customers with a smile. Say "please" and "thank you."

2. **Be Honest:** Be truthful about your product. If the eggs were laid 3 days ago, say so. Don't sell bad products. Honesty builds *trust*.
3. **Be Reliable and Punctual:** If you promise to deliver 50 chickens on Friday, deliver 50 chickens *on Friday*.
4. **Listen to Feedback:** If a customer complains, *listen* patiently. Apologize (even if it's not your fault) and offer a solution (e.g., "I'm sorry about that. Here is a replacement.").
5. **Add Small "Extras":** (e.g., "lagniappe"). If a customer buys 10 tomatoes, add one extra for free. This builds loyalty.

Evaluation:

1. What is a "marketing strategy"?
2. List the "4 Ps" of marketing.
3. What does the "P" for "Promotion" mean?
4. What is "customer relations"?
5. List two ways to maintain good customer relations.

Assignment:

1. List and *explain* the "4 Ps" of marketing in your own words.
2. What is the *most powerful* form of promotion for a small business?
3. Why is it important to be *honest* with customers?
4. Imagine a customer complains angrily that the eggs you sold them were bad. What is the *professional* way to handle this? (Write 2-3 sentences).
5. (Practical) Design a simple *brand name* and *label* for your imaginary farm (e.g., "Sunshine Veggies - Fresh & Clean").

WEEK 10: BUSINESS SIMULATION

Topic: Business Simulation

Content: Group projects and business idea development.

Performance Objectives: By the end of this lesson, students should be able to:

1. Work effectively as a group.
2. Brainstorm and select one viable agribusiness idea for a school-based project.
3. Develop a simple business plan for their idea (Name, Goal, Budget, Marketing).
4. Present their business idea to the class.
5. Apply the concepts of entrepreneurship, financial management, and marketing.



Lesson Content:

1. Goal of the Simulation

- This is a practical, group-based project that combines *all* the topics from this term (Weeks 1, 2, 3, 4, 6, 8, and 9).
- Each group will act as "agripreneurs" and develop a plan for a small, school-based business.

2. Group Task: Develop a Business Idea (The class is divided into groups of 4-5 students).

- **Step 1: Brainstorming (10 mins)**

- "What are the *needs* in our school?" (e.g., Students are hungry at break, teachers want fresh vegetables).
- "What *resources* do we have?" (e.g., A small plot of land for a garden, space for a rabbit hutch).
- **Ideas:**
 - *Idea A:* A small Ugu/Spinach garden (Horticulture).
 - *Idea B:* A small snack business (Processing - e.g., making and selling plantain chips or zobo).
 - *Idea C:* A small livestock farm (e.g., 5-10 snails, 2-3 rabbits).

- **Step 2: Create a Simple Business Plan (20 mins)**

- Each group must prepare a short presentation covering these 8 points:
2. **Business Name:** (e.g., "JSS2 Green Eagles Garden," "Speedy Snacks").
 3. **The Opportunity:** (What problem are you solving?).
 4. **The Product:** (What *exactly* are you selling?).
 5. **Target Customers (Place):** (Who will you sell to? Students? Teachers? Parents?).
 6. **Simple Budget (Expenditure):** (What 3-5 things do you need to *buy* to start, and what is the *total* cost?).
 7. **Pricing:** (How much will you sell one item for?).
 8. **Promotion:** (How will you *advertise*? e.g., "A poster in the hall," "Announce at assembly").

9. **Profit Goal:** (How much do you hope to make in one month?).

- **Step 3: Group Presentations (15 mins)**

- Each group selects a "CEO" to present their 2-minute business plan to the class (the "investors").
- The class listens and asks questions.
- The teacher provides feedback on the plan's feasibility (if it can work).

3. Example Plan (for "JSS2 Green Eagles Garden")

1. **Name:** JSS2 Green Eagles Garden
2. **Opportunity:** Teachers want fresh ugu for soup but the market is far.
3. **Product:** Fresh, organic Ugu and Spinach.
4. **Customers:** Teachers in the staff room and parents at closing time.
5. **Budget (Expenditure):**
 - Ugu seeds: N500
 - Spinach seeds: N500
 - Organic Manure: N1,000
 - **Total:** N2,000
6. **Price:** N100 per bunch.
7. **Promotion:** A poster on the staff room door and word of mouth.
8. **Profit Goal:** To make N5,000 profit by the end of term.

Evaluation: (This is a practical group assessment).

1. Did the group work together?
2. Was their business idea realistic for a school?
3. Did their business plan include a Budget and a Marketing idea?
4. How well did they present their idea?

Assignment:

1. Write down the 8-point business plan for *your* group's idea.

2. What was the *biggest challenge* your group faced in creating the plan?
3. Which *other* group had the best idea (in your opinion), and why?
4. If you had N5,000, which of the class business ideas would you *personally* invest in, and why?
5. What is the most important thing you learned from this project?



Week 11: REVISION

Topic: Revision

Content: Review of term's topics.

Performance Objectives: By the end of this lesson, students should be able to:

1. Verbally define all key terms from the term (e.g., Vegetable, Agribusiness, Entrepreneur, Budget, 4 Ps of Marketing).
2. Explain the key stages of vegetable production.
3. Differentiate between sorting, grading, and packaging.
4. Explain the importance of entrepreneurship.
5. Explain the difference between a budget, an expense record, and a sales record.

Lesson Content:

1. Summary of Term 3 Topics (The "Business of Horticulture" Term) This term focused on applying horticultural skills as a *business*.

- **Week 1 (Vegetable Production):** We classified vegetables (Leafy, Fruit, Root, Bulb) and noted their nutritional/economic importance. We reviewed the production stages (site selection, land prep, nursery/transplanting, maintenance, harvesting).
- **Week 2 (Post-Harvest):** We learned that horticultural crops are *perishable*.
 - **Sorting:** Removing the *bad* produce.
 - **Grading:** Classifying the *good* produce by size, colour, etc., to increase value.
 - **Packaging:** Using *crates* or *nets* to protect produce.
 - **Storage:** Using **refrigeration** (short-term) or **drying/canning/processing** (long-term).
- **Week 3 (Agribusiness):** We defined agribusiness as the *entire value chain* (Inputs → Production → Processing → Marketing). We identified career opportunities in each sector.

- **Week 4 (Entrepreneurship):** An **entrepreneur** (agripreneur) is a *risk-taker* who *starts a business* to solve a problem for *profit*.
 - **Qualities:** Innovative, Persistent, Hard-working.
 - **Importance:** Creates jobs, ensures food security, develops rural areas.
- **Week 6 (Enterprise Development):** We learned the 5 steps to start a business (Idea → Plan → Capital → Resources → Production). We saw how *Record-Keeping* (Expenses, Sales) is used to calculate **Profit**.
- **Week 8 (Financial Management):**
 - **Rule #1:** *Separate* personal money from business money.
 - **Budget:** A *plan* for future income and expenses.
- **Week 9 (Marketing):**
 - **The 4 Ps:** Product (quality), Price (cost + profit), Place (where to sell), Promotion (advertising).
 - **Customer Relations:** Being honest, polite, and reliable to build trust and *keep* customers.
- **Week 10 (Simulation):** We practiced creating a simple business plan for a school-based project.

2. Class Activity: Rapid-Fire Q&A

- "Ugu is what type of vegetable?" (Leafy)
- "Tomato is what type of vegetable?" (Fruit)
- "Removing bad tomatoes from the good ones is...?" (Sorting)
- "Separating good tomatoes by size (Large/Small) is...?" (Grading)
- "The best packaging for tomatoes is...?" (Plastic crate)
- "What do we call a person who starts a farm to make a profit?"
(Agripreneur/Entrepreneur)
- "A written 'roadmap' for your business is a...?" (Business Plan)
- "The money you use to start a business is...?" (Capital)
- "The formula for profit is...?" (Sales - Expenses)
- "What are the 4 Ps of Marketing?" (Product, Price, Place, Promotion)
- "Being honest and polite to customers is called...?" (Customer Relations)

Evaluation: This week's evaluation is the Third Term (Final) Examination.

Assignment:

1. Study all notes from the term.
2. Make a list of 10 key terms and their definitions (e.g., Agribusiness, Entrepreneur, Budget, Grading, 4 Ps).
3. What is the difference between "Sorting" and "Grading"?
4. Why must an entrepreneur separate business money from personal money?
5. List the 4 Ps of Marketing.

